Deep Learning Interview Questions and Answers

1. What is Deep Learning, and how does it differ from traditional Machine Learning?

Deep Learning is a subset of Machine Learning that uses artificial neural networks with multiple layers to learn patterns from large amounts of data.

The main difference is that traditional Machine Learning often requires manual feature extraction, whereas Deep Learning can automatically learn features from raw data.

- Machine Learning: Identifying spam emails using manually selected features.
- Deep Learning: Recognising faces in images automatically.

2. What is an Artificial Neural Network (ANN)?

An Artificial Neural Network (ANN) is a computing model inspired by the human brain. It consists of interconnected artificial neurons that learn patterns from data.

It is mainly used for classification, prediction and pattern recognition.

Example: Predicting house prices based on area, location and number of rooms.

3. What is a biological neuron, and how is it related to an artificial neuron?

A biological neuron is a nerve cell in the human brain that receives and transmits information.

An artificial neuron is inspired by this process. It receives inputs, processes them using weights and bias, and produces an output.

Example: Just as a biological neuron processes signals, an artificial neuron processes numerical inputs.

4. What are the main components of an artificial neuron?

The main components of an artificial neuron are:

1. Inputs: Data given to the neuron.
2. Weights: Importance assigned to each input.
3. Bias: An additional value used to adjust the result.
4. Summation: Calculates the weighted sum of inputs and bias.

5. Activation Function: Determines the final output of the neuron.

Formula:

$$Y = f(w_1 x_1 + w_2 x_2 + b)$$

Where:

- x = Input
- w = Weight
- b = Bias
- f = Activation function

5. What are input, hidden and output layers in a neural network?

A neural network mainly consists of three types of layers:

1. Input Layer: Receives the input data.
2. Hidden Layer: Processes the data and learns patterns.
3. Output Layer: Produces the final prediction or result.

Example: In a cat and dog classification model:

- Input Layer: Receives an image.
- Hidden Layers: Extract features such as ears, eyes and shapes.
- Output Layer: Predicts whether the image is a cat or dog.

6. What are weights and biases in a neural network?

Weights determine the importance of each input in a neural network.

Bias is an additional value that helps adjust the neuron's output.

Both weights and biases are updated during training to improve the model's predictions.

Example: If house size is more important for predicting house prices, the model may learn a higher weight for house size.

7. What is a forward pass in a neural network?

A forward pass is the process of passing input data through the neural network to generate an output or prediction.

The data moves from the input layer through the hidden layers to the output layer.

Example: Giving a house's area and location to a neural network and getting its predicted price.

8. What is a loss function, and why is it necessary?

A loss function measures the difference between the actual output and the predicted output of a model.

It helps the model understand how wrong its prediction is and guides the training process to reduce the error.

Example: If the actual house price is ₹50 lakh and the predicted price is ₹45 lakh, the loss function measures the prediction error.

Common loss functions:

- Mean Squared Error (MSE) – Regression.
- Binary Cross-Entropy – Binary Classification.
- Categorical Cross-Entropy – Multi-class Classification.

9. What is the difference between an epoch, a batch and an iteration?

These three terms describe how data is processed during neural network training.

1. Epoch: One complete pass through the entire training dataset.
2. Batch: A small group of training samples processed together.
3. Iteration: One update of the model's parameters using one batch.

Example:

Suppose we have 1,000 training samples and a batch size of 100.

- Total samples = 1,000
- Batch size = 100
- Iterations per epoch = 10
- 1 Epoch = 10 Iterations.

10. What are the advantages and limitations of Deep Learning compared with traditional ML algorithms?

Advantages:

1. Automatically learns features from data.
2. Handles complex problems.
3. Works well with large datasets.
4. Useful for image, speech and text processing.
5. Can achieve high accuracy in many complex tasks.

Limitations:

1. Requires large amounts of data.
2. Needs high computational power.
3. Training can take a long time.
4. Difficult to interpret how the model makes decisions.
5. Requires careful tuning of hyperparameters.

11. What is an activation function, and why is it necessary?

An activation function is a mathematical function applied to the output of a neuron. It decides whether and how strongly a neuron should be activated.

It is necessary because it introduces non-linearity into a neural network, allowing the network to learn complex patterns and relationships from data.

Without activation functions, a neural network cannot effectively learn complex patterns.

Examples:

- ReLU
- Sigmoid
- Tanh
- Softmax

12. What happens if a neural network uses no nonlinear activation functions?

If a neural network uses no nonlinear activation functions, it behaves like a linear model, even if it contains multiple hidden layers.

This means the network cannot learn complex patterns or relationships in the data.

Example:

A neural network may struggle to identify complex image patterns if it uses only linear operations.

Key Point: Nonlinear activation functions help neural networks learn complex relationships.

13. Explain the Sigmoid activation function and its limitations.

Sigmoid is an activation function that converts any input value into a value between 0 and 1.

It is mainly used for binary classification problems because its output can be interpreted as a probability.

Formula:

$$f(x) = \frac{1}{1+e^{-x}}$$

Example:

For a spam detection model:

- Output 0.9 means a 90% predicted probability of spam.
- Output 0.1 means a 10% predicted probability of spam.

Limitations:

1. Vanishing gradient problem: Gradients become very small for extreme input values, making learning slow.
2. Not zero-centred: Its outputs are always positive.
3. Can slow down training in deep neural networks.

14. What is the Tanh activation function, and how does it differ from Sigmoid?

Tanh (Hyperbolic Tangent) is an activation function that converts input values into a range between -1 and +1.

Unlike Sigmoid, which produces only positive values, Tanh produces both positive and negative values.

Formula:

$$f(x) = \frac{e^{x}-e^{-x}}{e^{x}+e^{-x}}$$

Difference between Sigmoid and Tanh:

Sigmoid	Tanh
Output range: 0 to 1	Output range: -1 to +1

Not zero-centred	Zero-centred
Mainly used for binary output	Historically used in hidden layers and recurrent networks
Suffers from vanishing gradients	Also suffers from vanishing gradients

Key Point: Tanh is zero-centred, whereas Sigmoid is not.

15. What is the ReLU activation function, and why is it widely used?

ReLU stands for Rectified Linear Unit. It is an activation function that returns 0 for negative inputs and the input value itself for positive inputs.

Formula:

$$f(x) = \max(0, x)$$

Example:

- Input = -5 → Output = 0
- Input = 5 → Output = 5
- Input = 10 → Output = 10

Why is it widely used?

1. Simple and computationally efficient.
2. Helps reduce the vanishing gradient problem for positive inputs.
3. Speeds up training.
4. Works well in many deep neural networks.

Key Point: ReLU is one of the most commonly used activation functions in hidden layers.

16. What is the Dying ReLU problem?

The Dying ReLU problem occurs when a ReLU neuron continuously produces an output of zero because its input remains negative.

In this situation, the gradient becomes zero, so the neuron stops learning and updating its weights.

Example:

If a neuron's input is always -5, ReLU produces 0. As a result, that neuron may stop contributing to the learning process.

Solution:

We can use Leaky ReLU instead of standard ReLU because it allows a small negative output.

17. What are Leaky ReLU, ELU and GELU?

Leaky ReLU, ELU and GELU are alternative activation functions designed to improve upon standard ReLU.

1. Leaky ReLU:

Leaky ReLU allows a small negative output instead of returning zero for all negative inputs.

This helps reduce the dying ReLU problem.

Example:

- Input = -5 → Output = -0.05 (if the negative slope is 0.01).
- Input = 5 → Output = 5.

2. ELU (Exponential Linear Unit):

ELU produces a smooth negative output for negative inputs and returns the input itself for positive values.

It helps maintain small negative outputs instead of completely removing them.

3. GELU (Gaussian Error Linear Unit):

GELU smoothly adjusts input values based on their magnitude rather than simply switching between zero and the input value.

It is widely used in Transformer-based models, including BERT.

Key Difference:

Function	Main characteristic

Leaky ReLU	Allows a small negative output
ELU	Uses a smooth exponential curve for negative inputs
GELU	Smoothly scales inputs based on their values

18. What is the Softmax function, and when is it used?

Softmax is an activation function that converts multiple numerical outputs into probability values between 0 and 1, where all the probabilities add up to 1.

It is mainly used in the output layer of multi-class classification models.

Formula:

$$Softmax(x_i) = \frac{e^{x_i}}{\sum_j e^{x_j}}$$

Example:

Suppose a model identifies an image containing one of three animals.

Animal	Probability
Cat	0.70
Dog	0.20
Horse	0.10

The model predicts Cat because it has the highest probability.

Key Point: Softmax is commonly used for multi-class classification.

19. How do you choose an activation function for the hidden and output layers?

The choice of activation function depends on the type of problem and the expected output.

For Hidden Layers:

- ReLU – Common default choice.
- Leaky ReLU – Useful when addressing the dying ReLU problem.
- GELU – Commonly used in Transformer architectures.
- Tanh – Useful in some recurrent neural network architectures.

For Output Layers:

- Sigmoid – Binary classification.
- Softmax – Multi-class classification.
- Linear – Regression problems where the model predicts continuous values.

Example:

For a house price prediction model:

- Hidden layers: ReLU.
- Output layer: Linear activation.

For a cat, dog and horse classification model:

- Hidden layers: ReLU.
- Output layer: Softmax.

20. What is the difference between a Single-Layer Perceptron and a Multilayer Perceptron (MLP)?

A Single-Layer Perceptron is the simplest type of neural network. It has an input layer and an output layer but no hidden layer.

A Multilayer Perceptron (MLP) contains one or more hidden layers between the input and output layers. It can learn more complex patterns using nonlinear activation functions.

Difference:

Single-Layer Perceptron	Multilayer Perceptron
No hidden layers	One or more hidden layers
Simple architecture	More complex architecture
Learns linearly separable patterns	Can learn nonlinear patterns
Limited learning capability	More powerful learning capability
Requires less computation	Requires more computation

Example:

- Single-Layer Perceptron: Simple classification of linearly separable data.
- MLP: Handwritten digit recognition.

Key Point: A single-layer perceptron can solve simple linear problems, whereas an MLP can solve more complex nonlinear problems.

21. What is Forward Propagation, and how are predictions calculated?

Forward propagation is the process of passing input data through a neural network to calculate the final prediction.

The input data passes through the input layer, hidden layers and output layer. Each neuron performs calculations using weights and biases, followed by an activation function.

The final output is the model's prediction.

Example:

In a house price prediction model:

1. Input: House area and number of rooms.

2. Hidden layers: Process the input features.
3. Output layer: Predicts the house price.

Key Point: Forward propagation moves data from the input layer to the output layer to generate predictions.

22. What is Backpropagation, and why is it important?

Backpropagation is the process of calculating how much each weight and bias contributes to the prediction error.

It works backwards from the output layer towards the input layer and calculates gradients using the chain rule.

These gradients help the optimisation algorithm update the weights and biases to reduce the loss.

Example:

Suppose a model predicts a house price of ₹40 lakh, but the actual price is ₹50 lakh.

Backpropagation calculates how the network's parameters contributed to this error so they can be adjusted during training.

Key Point: Forward propagation calculates predictions, whereas backpropagation calculates gradients to help improve predictions.

23. How does the Chain Rule help calculate gradients in neural networks?

The chain rule is a mathematical rule used to calculate the derivative of a function that depends on another function.

In neural networks, the output of one layer becomes the input to the next layer. Therefore, the chain rule helps calculate how changes in earlier weights affect the final loss.

It allows gradients to be calculated layer by layer during backpropagation.

Example:

Consider a neural network with three layers.

If the final prediction is incorrect, the chain rule helps calculate how much each layer and its weights contributed to the error.

Key Point: The chain rule helps calculate gradients throughout the neural network during backpropagation.

24. What is a Gradient in the context of Deep Learning?

A gradient measures how much the loss function changes when we change the weights or biases of a neural network.

It indicates the direction and rate of change of the loss.

Gradients are used by optimisation algorithms to update the model's parameters and reduce the prediction error.

Example:

- Positive gradient: Increasing the weight increases the loss locally.
- Negative gradient: Increasing the weight decreases the loss locally.
- Zero gradient: The loss has no local change with respect to that parameter.

Key Point: Gradients tell the model how its parameters should change to reduce the loss.

25. How are Weights and Biases updated during training?

Weights and biases are updated using an optimisation algorithm such as Gradient Descent.

The algorithm uses the gradients calculated during backpropagation to adjust the parameters and reduce the loss.

The learning rate controls how large each update should be.

Weight update formula:

$$w_{new} = w_{old} - \alpha \frac{\partial L}{\partial w}$$

Where:

- w = Weight
- α = Learning rate
- L = Loss function
- $\frac{\partial L}{\partial w}$ = Gradient of the loss with respect to the weight.

Example:

If a weight is 0.8, the learning rate is 0.1 and the gradient is 0.5:

$$w_{new} = 0.8 - (0.1 \times 0.5)$$

$$w_{new} = 0.75$$

The new weight becomes 0.75.

Key Point: Weights and biases are updated to minimise the loss and improve the model's predictions.

26. What is the difference between a Loss Function and an Optimisation Algorithm?

A loss function measures how far the model's prediction is from the actual output.

An optimisation algorithm uses the gradients to update the weights and biases to reduce the loss.

Example:

Suppose a house price prediction model predicts ₹40 lakh, but the actual price is ₹50 lakh.

- Loss Function: Measures the prediction error.
- Optimisation Algorithm: Adjusts the model's weights and biases to reduce that error.

Difference:

Loss Function	Optimisation Algorithm
Measures prediction error	Reduces prediction error
Calculates the loss value	Updates weights and biases
Evaluates model performance during training	Controls parameter updates
Examples: MSE, Cross-Entropy	Examples: SGD, Adam, RMSprop

Key Point: The loss function tells us how wrong the prediction is, while the optimisation algorithm helps the model correct it.

27. What is the Vanishing Gradient Problem?

The vanishing gradient problem occurs when gradients become extremely small during backpropagation, especially in deep neural networks.

As a result, the weights in earlier layers receive very small updates and learn very slowly or may stop learning effectively.

It commonly occurs with activation functions such as Sigmoid and Tanh when they operate in their saturated regions.

Example:

In a neural network with many hidden layers, the gradients may become smaller as they move backwards towards the earlier layers.

This makes training difficult.

Solutions:

1. Use ReLU or suitable alternatives.
2. Use proper weight initialisation.
3. Use Batch Normalisation where appropriate.
4. Use residual connections in deep networks.

Key Point: Vanishing gradients make it difficult for earlier layers to learn because their gradients become too small.

28. What is the Exploding Gradient Problem?

The exploding gradient problem occurs when gradients become extremely large during backpropagation.

As a result, weights and biases receive very large updates, making training unstable.

It commonly occurs in very deep neural networks and recurrent neural networks.

Example:

Suppose a weight is 0.5, but a very large gradient causes it to increase or decrease drastically during an update.

This can make the model produce unstable predictions.

Solutions:

1. Use gradient clipping.
2. Apply proper weight initialisation.
3. Use suitable learning rates.
4. Use appropriate network architectures.

Key Point: Exploding gradients cause excessively large parameter updates and make training unstable.

29. How can Gradient Clipping help stabilise training?

Gradient clipping is a technique used to prevent gradients from becoming excessively large during neural network training.

It limits the gradient values to a specified threshold before updating the weights.

This helps prevent large parameter updates and makes training more stable.

Example:

Suppose the gradient value is 100, but the clipping threshold is 5.

The gradient is restricted to the specified limit instead of using the excessively large value.

Why is it used?

1. Prevents exploding gradients.
2. Improves training stability.
3. Helps avoid excessively large weight updates.

Key Point: Gradient clipping controls large gradients to make neural network training more stable.

30. How would you calculate the output of a simple neural network with given weights, biases and activation functions?

To calculate the output of a simple neural network, we perform forward propagation step by step.

We use the given inputs, weights, biases and activation functions to calculate the final output.

Example:

Consider a simple neural network with two input neurons and one output neuron.

Network structure

Illustrative single-neuron network with two inputs.

Given values:

- Input 1 = 2
- Input 2 = 3
- Weight 1 = 0.5
- Weight 2 = 0.3
- Bias = 0.2
- Activation function = ReLU

Step 1: Calculate the weighted sum.

$$z = (x_1 w_1) + (x_2 w_2) + b$$

Substitute the values:

$$z = (2 \times 0.5) + (3 \times 0.3) + 0.2$$

$$z = 1 + 0.9 + 0.2 = 2.1$$

Step 2: Apply the activation function.

ReLU formula:

$$f(z) = \max(0, z)$$

$$f(2.1) = \max(0, 2.1) = 2.1$$

Final Output = 2.1

Key Point: First calculate the weighted sum, add the bias, and then apply the activation function to get the final output.

31. What is Mean Squared Error (MSE), and when is it used?

Mean Squared Error (MSE) is a loss function used to measure the average squared difference between actual values and predicted values.

It calculates the error by subtracting the predicted value from the actual value, squaring the difference and calculating the average.

Formula:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Where:

- y_i = Actual value.
- $\hat{y}_i$ = Predicted value.
- n = Number of samples.

Example:

Suppose:

- Actual value = 100
- Predicted value = 90

Squared error:

$$(100 - 90)^2 = 100$$

When is it used?

MSE is mainly used in regression problems, such as house price prediction and sales forecasting.

Key Point: MSE gives more importance to larger errors because it squares the differences.

32. What is Mean Absolute Error (MAE), and how does it differ from MSE?

Mean Absolute Error (MAE) is a loss function that calculates the average absolute difference between actual and predicted values.

Unlike MSE, it does not square the errors. It simply takes their absolute values.

Formula:

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|$$

Example:

Suppose:

- Actual value = 100
- Predicted value = 90

Absolute error:

$$|100 - 90| = 10$$

Difference between MAE and MSE:

MAE	MSE
Uses absolute errors	Uses squared errors
Less sensitive to outliers	More sensitive to outliers
Easier to interpret in the original unit	Expressed in squared units
Commonly used in regression	Commonly used in regression

Key Point: MAE treats errors proportionally, whereas MSE penalises larger errors more heavily.

33. What is Binary Cross-Entropy Loss?

Binary Cross-Entropy is a loss function used in binary classification problems where there are only two possible classes, such as Yes/No or 0/1.

It measures the difference between the actual class and the predicted probability.

It is commonly used with the Sigmoid activation function in the output layer.

Example:

Consider an email spam detection model:

- Class 1 = Spam
- Class 0 = Not Spam

If the actual class is Spam and the model predicts a probability of 0.90, the prediction is good and the loss is low.

If the model predicts 0.10, the prediction is poor and the loss is high.

When is it used?

- Spam detection.
- Fraud detection.
- Disease classification.
- Customer churn prediction.

Key Point: Binary Cross-Entropy measures prediction error when there are two possible classes.

34. What is Categorical Cross-Entropy, and when is Sparse Categorical Cross-Entropy preferable?

Categorical Cross-Entropy is a loss function used in multi-class classification problems where the target labels are represented using one-hot encoding.

Sparse Categorical Cross-Entropy performs the same type of loss calculation but uses integer class labels instead of one-hot encoded labels.

Example:

Suppose we are building an animal classification model with three classes:

- Cat = 0
- Dog = 1

- Horse = 2

Categorical Cross-Entropy:

The actual label is represented using one-hot encoding.

For Dog:

```
[0, 1, 0]
```

Sparse Categorical Cross-Entropy:

The actual label is represented using an integer.

For Dog:

```
1
```

Difference:

Categorical Cross-Entropy	Sparse Categorical Cross-Entropy
Uses one-hot encoded labels	Uses integer labels
Example: [0, 1, 0]	Example: 1
Requires one-hot encoding	Does not require one-hot encoding
Used for multi-class classification	Used for multi-class classification

Key Point: Both are used for multi-class classification. The main difference is how the actual labels are represented.

35. What is Gradient Descent, and how does it minimise a loss function?

Gradient Descent is an optimisation algorithm used to minimise the loss function by updating the weights and biases of a neural network.

It calculates the gradients of the loss and adjusts the parameters in the opposite direction of the gradient.

This process is repeated until the loss is reduced.

Example:

Suppose a model predicts a house price incorrectly.

1. The model makes a prediction.
2. The loss function calculates the error.
3. Gradients are calculated through backpropagation.
4. Gradient Descent updates the weights and biases.
5. The model makes another prediction.

This process continues during training to reduce the error.

Key Point: Gradient Descent minimises the loss by repeatedly updating model parameters.

36. What is the difference between Batch, Stochastic and Mini-Batch Gradient Descent?

These are three types of Gradient Descent that differ in how many training samples are used to calculate gradients and update the model parameters.

1. Batch Gradient Descent:

It uses the entire training dataset to calculate the gradient before updating the weights.

Example: If we have 1,000 samples, all 1,000 samples are used for one update.

2. Stochastic Gradient Descent (SGD):

It uses only one training sample to calculate the gradient and update the weights.

Example: If we have 1,000 samples, the model updates its weights after processing each sample.

3. Mini-Batch Gradient Descent:

It divides the training dataset into smaller batches and updates the weights after processing each batch.

Example: If we have 1,000 samples and a batch size of 100, the model processes 100 samples per update.

Difference:

Batch GD	Stochastic GD	Mini-Batch GD
Uses the entire dataset	Uses one sample	Uses a small group
One update per epoch	Many updates per epoch	Multiple updates per epoch
More memory required	Less memory required	Balanced memory usage
Stable gradient	Noisy gradient	More balanced gradient

Key Point: Mini-Batch Gradient Descent is widely used in Deep Learning because it provides a good balance between computational efficiency and stable updates.

37. What is the Learning Rate, and what happens when it is too high or too low?

The Learning Rate is a hyperparameter that controls how much the weights and biases are updated during each training step.

It determines the size of the update made by the optimisation algorithm.

Example:

Suppose the learning rate is 0.1.

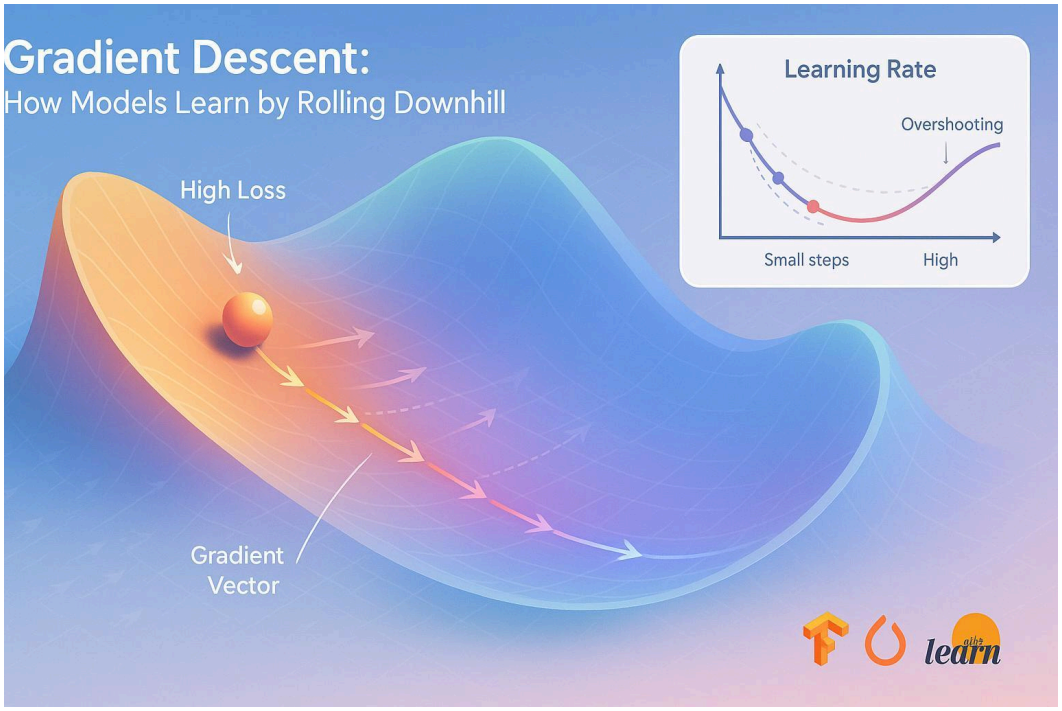
The optimiser uses this value to control how much the model's weights change based on the calculated gradients.

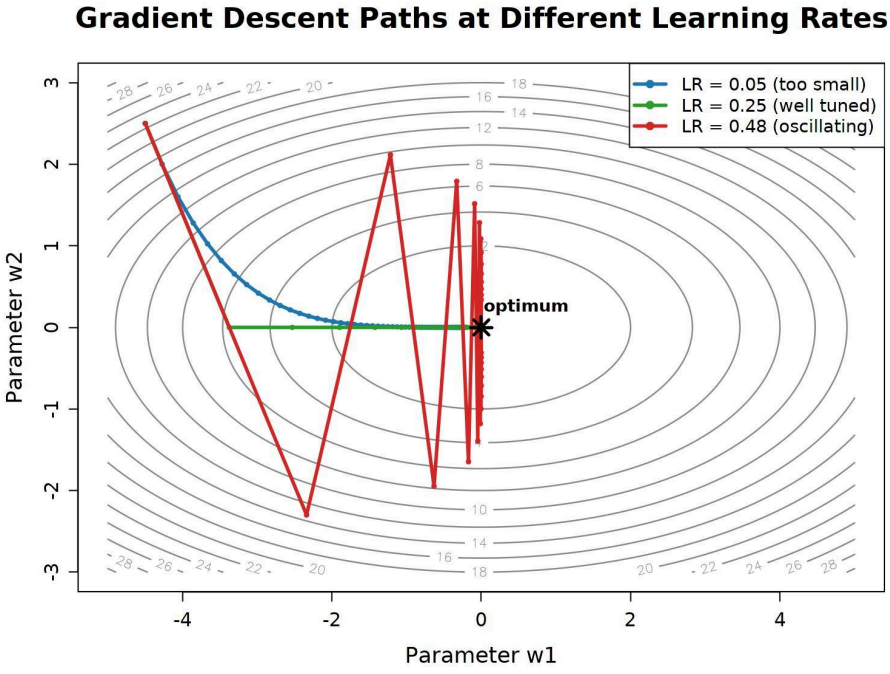
What happens if the learning rate is too high?

- The model may skip the optimal solution.
- Training may become unstable.
- The loss may increase instead of decreasing.

What happens if the learning rate is too low?

- Training becomes very slow.
- The model requires more iterations.
- It may take a long time to reach a good solution.





38. How does the Adam Optimiser work, and how does it differ from SGD?

Adam stands for Adaptive Moment Estimation. It is an optimisation algorithm that combines the ideas of Momentum and RMSprop.

It uses previous gradients and squared gradients to adjust the learning rate for each model parameter.

This helps the model update its weights efficiently during training.

Example:

Suppose we are training a deep neural network.

Adam automatically adjusts parameter updates based on the gradients, helping the model learn efficiently without manually adjusting the learning rate for every parameter.

Difference between Adam and SGD:

Adam	SGD
Adapts learning rates for individual parameters	Uses a common learning rate
Uses moving averages of gradients and squared gradients	Uses the current gradient
Often converges faster initially	May require more tuning
Uses more memory	Uses less memory

Key Point: Adam combines Momentum and RMSprop ideas to provide adaptive parameter updates. It is a popular optimiser for Deep Learning.

39. What are Momentum and RMSprop, and how do they improve optimisation?

Momentum and RMSprop are optimisation techniques that improve the training process of neural networks.

1. Momentum:

Momentum uses information from previous gradients to determine the current weight update.

It helps the optimiser move faster in the correct direction and reduces unnecessary fluctuations during training.

Example:

Imagine a ball rolling down a hill. It gains speed as it moves in the same direction. Momentum works in a similar way during optimisation.

2. RMSprop (Root Mean Square Propagation):

RMSprop adjusts the learning rate for each parameter based on the recent history of squared gradients.

It helps control large updates and improves training stability.

Difference:

Momentum	RMSprop
Uses previous gradients	Uses previous squared gradients
Helps accelerate updates	Adapts learning rates
Reduces fluctuations	Helps stabilise updates
Adds momentum to updates	Scales updates based on gradient magnitude

Key Point: Momentum helps accelerate learning, whereas RMSprop adjusts learning rates to improve stability.

40. What is a Learning-Rate Scheduler, and why might you use one?

A Learning-Rate Scheduler is a technique that automatically changes the learning rate during model training.

Instead of keeping the same learning rate throughout training, it adjusts the learning rate according to a predefined strategy or training performance.

Usually, a higher learning rate is used initially, and it is reduced as training progresses.

Example:

Suppose we start training a neural network with a learning rate of 0.01.

As training progresses, the scheduler reduces it to 0.001 and then 0.0001.

This helps the model make larger improvements initially and smaller adjustments near a good solution.

Why is it used?

1. Helps improve training stability.
2. Allows faster learning initially.
3. Helps the model converge towards a better solution.
4. Reduces unnecessary large updates near the minimum.

Common Learning-Rate Scheduling Techniques:

- Step Decay – Reduces the learning rate after a fixed number of epochs.
- Exponential Decay – Gradually reduces the learning rate exponentially.
- ReduceLROnPlateau – Reduces the learning rate when the monitored metric stops improving.
- Cosine Annealing – Changes the learning rate following a cosine curve.

41. What is Overfitting in a Neural Network, and how can you detect it?

Overfitting occurs when a neural network learns the training data too well, including unnecessary details and noise, but performs poorly on new, unseen data.

In this situation, the model achieves high accuracy on training data but low accuracy on validation or test data.

Example:

Suppose a model achieves:

- Training accuracy = 98%
- Validation accuracy = 70%

This indicates that the model may be overfitting.

How can you detect it?

1. Training accuracy is very high.
2. Validation accuracy is comparatively low.
3. Training loss decreases, but validation loss starts increasing.

How can you prevent it?

- Use Dropout.
- Apply L1 or L2 regularisation.
- Use Early Stopping.
- Increase training data.
- Reduce model complexity.

Key Point: Overfitting happens when a model performs well on training data but poorly on unseen data.

42. What is Underfitting, and how can you address it?

Underfitting occurs when a neural network is too simple to learn the important patterns from the training data.

As a result, the model performs poorly on both training and validation data.

Example:

Suppose a model achieves:

- Training accuracy = 60%
- Validation accuracy = 58%

This indicates that the model may be underfitting.

How can you address it?

1. Increase the number of hidden layers or neurons.
2. Train the model for more epochs.
3. Reduce excessive regularisation.
4. Use better input features.
5. Choose a more suitable model architecture.

Key Point: Underfitting happens when a model fails to learn the important patterns in the data.

43. What is the Bias–Variance Trade-off in Deep Learning?

The Bias–Variance Trade-off describes the balance between two types of model errors: bias and variance.

1. Bias:

Bias occurs when a model makes too many simplifying assumptions and fails to learn important patterns.

High bias usually leads to underfitting.

2. Variance:

Variance occurs when a model is too sensitive to the training data and learns unnecessary details or noise.

High variance usually leads to overfitting.

Example:

- High Bias: A simple model cannot learn complex patterns.
- High Variance: A complex model memorises training data but performs poorly on new data.

Key Point: The goal is to balance bias and variance so that the model performs well on both training and unseen data.

44. What is Dropout, and how does it reduce Overfitting?

Dropout is a regularisation technique used in neural networks to reduce overfitting.

During training, it randomly disables a percentage of neurons in a layer. This prevents the network from depending too much on specific neurons.

As a result, the network learns more general patterns.

Example:

Suppose a neural network has 100 neurons and a dropout rate of 0.2.

During each training step, approximately 20% of the neurons are randomly disabled.

Why is it used?

1. Reduces overfitting.
2. Prevents excessive dependency on individual neurons.
3. Improves generalisation to unseen data.

Key Point: Dropout randomly disables neurons during training to help prevent overfitting.

45. What is Batch Normalisation, and why is it useful?

Batch Normalisation is a technique used to normalise the inputs or activations of a neural network layer using the mean and variance calculated from the current mini-batch during training.

It helps maintain more consistent values throughout the network.

Why is it useful?

1. Helps stabilise training.
2. Can allow higher learning rates.
3. Often improves training speed.
4. Can provide a small regularisation effect.

Example:

Suppose the values entering a neural network layer have very different scales.

Batch Normalisation adjusts these values to make training more stable.

Key Point: Batch Normalisation normalises layer activations using mini-batch statistics to improve training stability.

46. What is Layer Normalisation, and how does it differ from Batch Normalisation?

Layer Normalisation is a technique that normalises the activations across the features of each individual sample.

Unlike Batch Normalisation, it does not depend on the statistics of the entire mini-batch.

It is widely used in Transformer architectures.

Difference:

Batch Normalisation	Layer Normalisation
Uses mini-batch statistics	Uses individual sample statistics

Normalises across the batch dimension for each feature	Normalises across features within each sample
Depends on batch statistics during training	Works independently of other samples
Common in CNNs	Common in Transformers

Example:

- Batch Normalisation: Commonly used in image classification CNNs.
- Layer Normalisation: Commonly used in BERT and Transformer models.

Key Point: Batch Normalisation uses batch statistics, whereas Layer Normalisation normalises each sample independently across its features.

47. What is L1 and L2 Regularisation in Neural Networks?

L1 and L2 Regularisation are techniques used to reduce overfitting by adding a penalty to the loss function based on the model's weights.

1. L1 Regularisation (Lasso):

L1 adds the absolute values of the weights to the loss function.

It encourages some weights to become exactly zero, which can make the model simpler.

2. L2 Regularisation (Ridge):

L2 adds the squared values of the weights to the loss function.

It discourages very large weights by pushing them towards zero.

Difference:

L1 Regularisation	**L2 Regularisation**
Uses absolute weight values	Uses squared weight values

Can make weights exactly zero	Usually makes weights small
Encourages sparse models	Encourages smaller weights
Useful for feature selection	Useful for controlling large weights

Key Point: L1 encourages sparse weights, whereas L2 discourages excessively large weights.

48. What is Early Stopping, and how does it work?

Early Stopping is a regularisation technique that stops model training when the validation performance stops improving.

Instead of training for a fixed number of epochs, the model monitors validation loss or another validation metric.

If the performance does not improve for a specified number of epochs, training is stopped.

Example:

Suppose the validation loss decreases until epoch 15 but starts increasing afterwards.

If the model continues training, it may start overfitting.

Early Stopping can stop training and restore the model weights from the best-performing epoch.

Why is it used?

1. Prevents overfitting.
2. Saves training time.
3. Helps select a better-performing model.

Key Point: Early Stopping stops training when validation performance stops improving.

49. How do Training and Validation Loss Curves help diagnose Model Performance?

Training and validation loss curves are graphs that show how the model's error changes during training.

They help us understand whether the model is learning properly, overfitting or underfitting.

1. Training Loss:

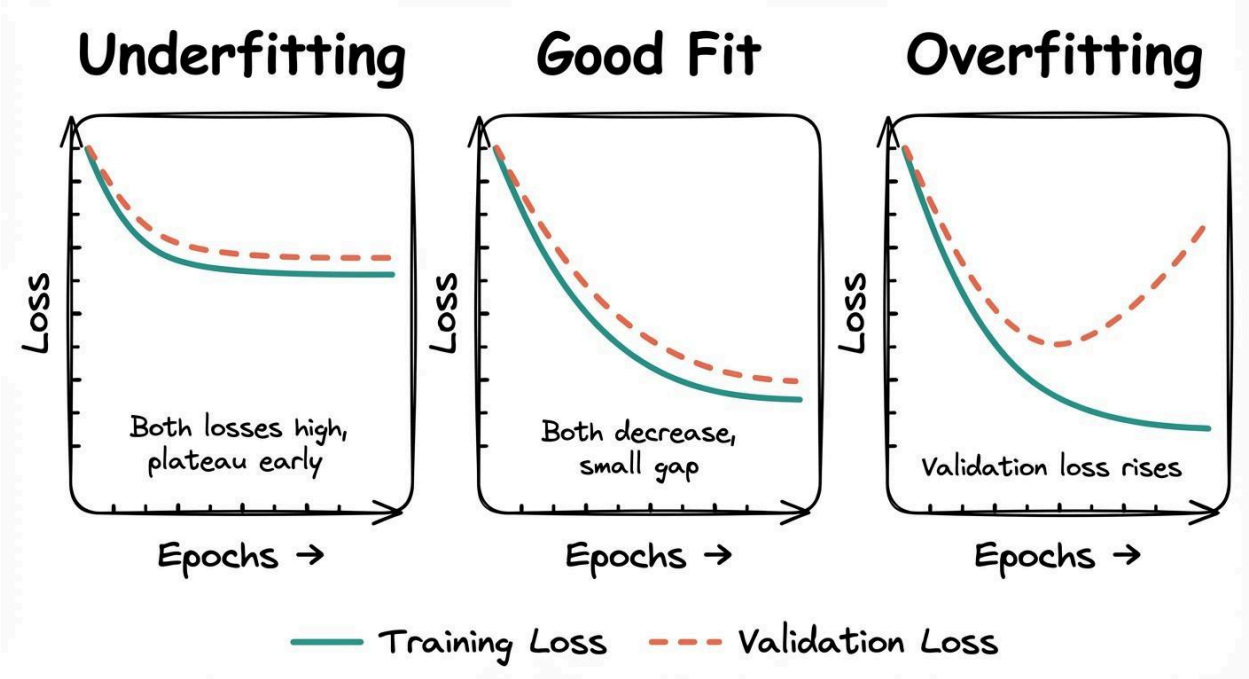
Measures the model's error on the training dataset.

2. Validation Loss:

Measures the model's error on a separate validation dataset that is not used to update the model's weights.

Common patterns:

Overfitting

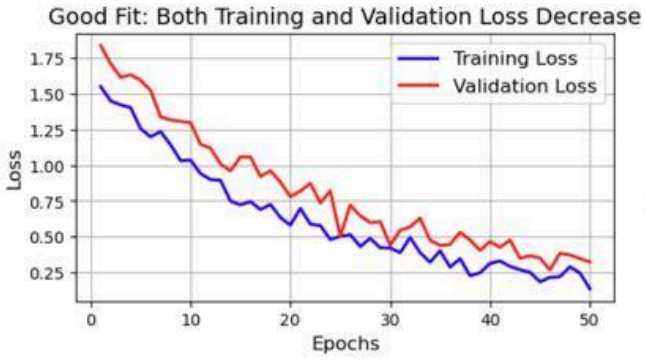


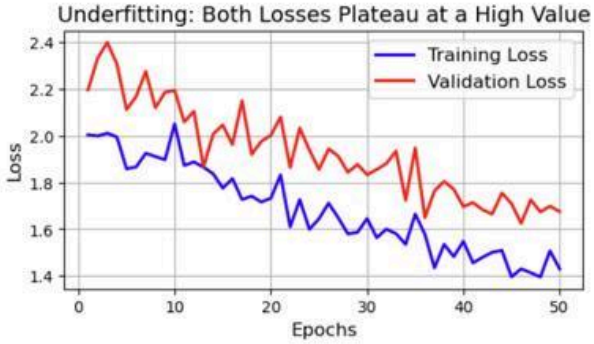
Training loss decreases, but validation loss starts increasing.

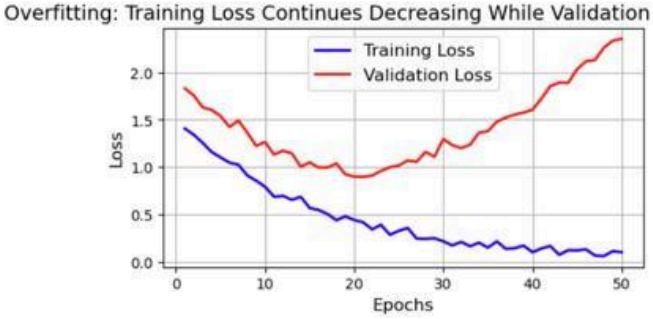
Underfitting

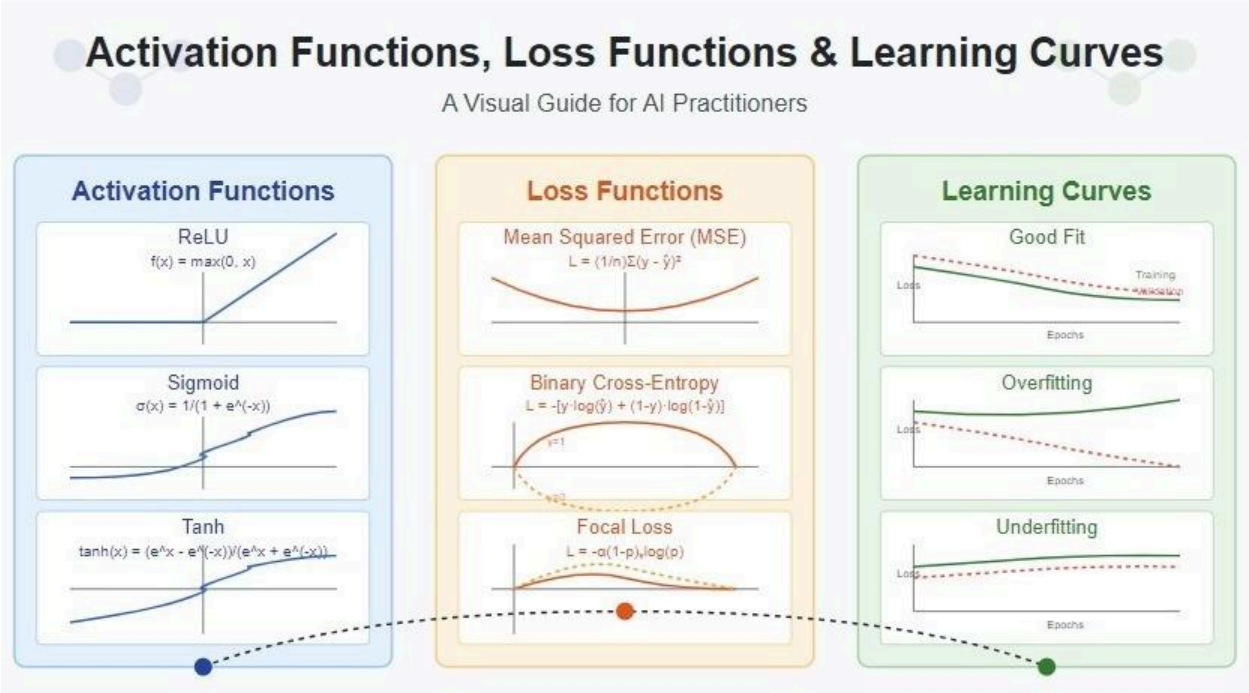
Both training and validation loss remain relatively high.

Good Generalisation









Both losses decrease and remain reasonably close.

Key Point: Training and validation loss curves help identify overfitting, underfitting and good model generalisation.

50. A model achieves excellent Training Accuracy but poor Validation Accuracy. How would you investigate and improve it?

This situation usually indicates overfitting. It means the model has learned the training data well but cannot generalise effectively to unseen data.

I would investigate the model and apply suitable techniques to improve its performance.

Steps to improve the model:

1. Check the dataset: Make sure the training and validation data are correctly prepared and representative.
2. Use Dropout: Randomly disable neurons during training to reduce overfitting.
3. Apply Regularisation: Use L1 or L2 regularisation to control model complexity.
4. Use Early Stopping: Stop training when validation performance stops improving.
5. Increase Training Data: Collect more data or use data augmentation where appropriate.
6. Reduce Model Complexity: Reduce the number of layers or neurons if the model is unnecessarily complex.
7. Check Data Leakage: Make sure information from the validation dataset is not accidentally used during training.

Example:

Suppose:

- Training accuracy = 99%
- Validation accuracy = 65%

I would first check the loss curves and data split, then investigate overfitting and apply suitable regularisation techniques.

Key Point: High training accuracy with poor validation accuracy usually indicates overfitting, so the goal is to improve the model's ability to generalise.

51. What is a Convolutional Neural Network (CNN), and why is it useful for image processing?

A Convolutional Neural Network (CNN) is a type of Deep Learning model mainly used for processing and analysing images.

It automatically learns important features from images, such as edges, shapes, textures and objects.

Unlike traditional neural networks, CNNs use convolutional layers to identify spatial patterns in images.

Example:

In a cat and dog classification model:

- CNN identifies edges and basic shapes.
- It learns features such as eyes, ears and fur.
- It uses these features to identify whether the image contains a cat or a dog.

Why is CNN useful?

1. Automatically extracts image features.
2. Detects patterns in images.
3. Requires fewer parameters than a fully connected network for typical image inputs.
4. Works well for image classification and object detection.

Key Point: CNNs are mainly used for image processing because they automatically learn visual features from images.

52. What is a Convolution Operation in a CNN?

Convolution is the process of applying a small filter or kernel over an input image to identify important patterns.

The filter moves across the image and performs mathematical calculations on small sections of pixels.

The result is a new representation called a feature map.

Example:

Suppose we give an image of a cat to a CNN.

A convolution operation can help identify:

- Edges.
- Curves.
- Shapes.
- Textures.

These features help the network recognise objects in the image.

Key Point: Convolution applies filters to images to extract useful features.

53. What are Filters or Kernels in a Convolutional Layer?

Filters, also called kernels, are small matrices of learnable values used in CNNs to detect specific patterns in images.

During convolution, these filters move across the input image and calculate values to identify features.

Different filters can learn to detect different patterns.

Example:

- One filter may detect vertical edges.
- Another filter may detect horizontal edges.
- Another filter may detect textures or shapes.

Common kernel sizes:

- 3 × 3
- 5 × 5
- 7 × 7

The filter values are learned automatically during model training.

Key Point: Filters are small learnable matrices that help CNNs extract different features from images.

54. What are Stride and Padding, and how do they affect Output Dimensions?

Stride and padding are two important parameters used in convolutional layers.

1. Stride:

Stride determines how many pixels the filter moves at each step across the input image.

Example:

- Stride = 1: Filter moves one pixel at a time.
- Stride = 2: Filter moves two pixels at a time.

A larger stride generally produces a smaller output feature map.

2. Padding:

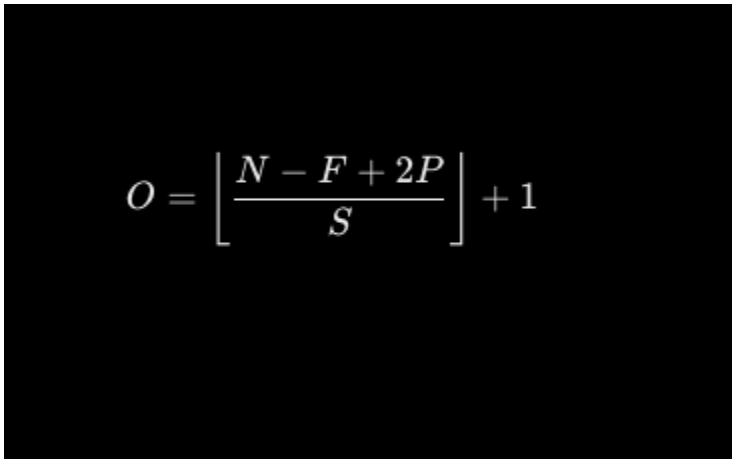
Padding means adding extra pixels, usually zeros, around the borders of an image before applying convolution.

It helps control the output dimensions and preserve information near the image boundaries.

Types of Padding:

- Valid Padding: No extra pixels are added.
- Same Padding: Padding is added to maintain the same spatial dimensions when stride is 1.

Output Dimension Formula:



$$O = \left\lfloor \frac{N - F + 2P}{S} \right\rfloor + 1$$

Where:

- N = Input size.
- F = Filter size.
- P = Padding.
- S = Stride.
- O = Output size.

Example:

Suppose:

- Input size = 32 × 32
- Filter size = 3 × 3
- Padding = 0
- Stride = 1

Output size:

$$O = \frac{32-3+0}{1} + 1 = 30$$

Final output = 30 × 30.

Key Point: Stride controls the filter movement, while padding controls the image boundaries and output dimensions.

55. What is a Feature Map?

A feature map is the output produced after applying a filter to an input image through a convolution operation.

It represents the patterns or features detected by that filter.

Each filter produces its own feature map.

Example:

When processing a cat image:

- One feature map may highlight edges.
- Another may highlight textures.
- Another may highlight shapes.

These feature maps help the CNN understand the image.

Key Point: A feature map represents the features detected by a convolutional filter.

56. What is Pooling, and what are Max Pooling and Average Pooling?

Pooling is a technique used in CNNs to reduce the spatial dimensions of feature maps while retaining important information.

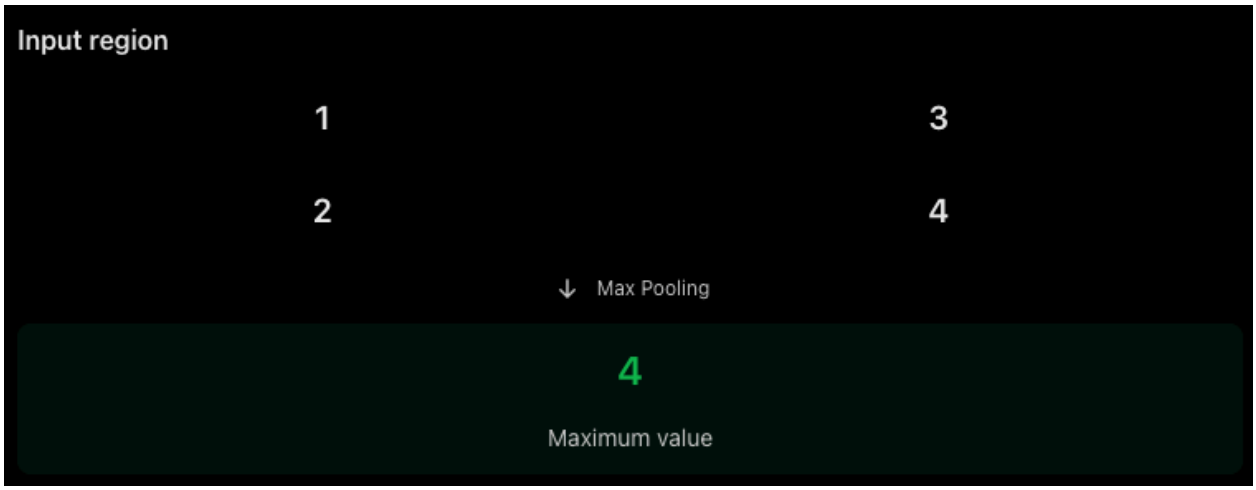
It reduces the computational requirements and helps make the network less sensitive to small changes in feature positions.

1. Max Pooling:

Max Pooling selects the maximum value from a small region of the feature map.

Example:

Consider a 2 × 2 region:



2. Average Pooling:

Average Pooling calculates the average value of a small region.

Using the same example:

$$\frac{1+3+2+4}{4} = 2.5$$

The output is 2.5.

Difference:

Max Pooling	Average Pooling
Selects the maximum value	Calculates the average value
Preserves strong features	Preserves average information
Commonly used in CNNs	Used when average information is useful

Key Point: Pooling reduces feature map dimensions. Max Pooling selects the maximum value, while Average Pooling calculates the average.

57. What is the Difference Between Convolutional and Fully Connected Layers?

A Convolutional Layer extracts important features from images using filters or kernels.

A Fully Connected Layer connects every neuron in one layer to every neuron in the next layer. It uses the extracted features to make the final prediction.

Difference:

Convolutional Layer	Fully Connected Layer
Uses filters or kernels	Uses connections between all neurons
Extracts image features	Uses learned features for prediction
Preserves spatial structure	Usually operates on flattened features
Shares weights across image locations	Has separate weights for each connection
Commonly used in feature extraction	Commonly used in classification

Example:

In a cat and dog classification model:

- Convolutional Layer: Extracts eyes, ears, shapes and textures.
- Fully Connected Layer: Uses these features to predict Cat or Dog.

Key Point: Convolutional layers extract features, whereas fully connected layers use those features to make predictions.

58. What is the Receptive Field of a CNN?

The receptive field is the region of the original input image that influences a particular neuron in a CNN.

In simple terms, it tells us how much of the original image a neuron can observe.

As information passes through multiple convolutional and pooling layers, the receptive field generally becomes larger.

Example:

In the first convolutional layer, a neuron may observe a small 3 × 3 region of the image.

In deeper layers, a neuron may observe a much larger region.

This helps the network learn both small details and larger patterns.

Key Point: The receptive field determines how much of the original image a neuron can observe.

59. What is the Difference Between a Standard Convolution and a Depthwise-Separable Convolution?

Standard Convolution applies filters across all input channels to extract features and combine information from different channels.

Depthwise-Separable Convolution divides this operation into two steps to reduce computational cost.

1. Standard Convolution:

It performs spatial feature extraction and channel combination in a single operation.

2. Depthwise-Separable Convolution:

It performs two separate operations:

- Depthwise Convolution: Applies a separate filter to each input channel.
- Pointwise Convolution: Uses 1 × 1 filters to combine information from different channels.

Difference:

Standard Convolution	Depthwise-Separable Convolution

Single convolution operation	Two separate operations
More parameters	Fewer parameters
Higher computational cost	Lower computational cost
Extracts and combines features together	Separates spatial filtering and channel mixing

Example:

Depthwise-Separable Convolution is used in MobileNet architectures, which are designed for efficient image processing on mobile and resource-constrained devices.

Key Point: Depthwise-Separable Convolution reduces parameters and computational cost while maintaining useful feature extraction.

60. How would you design a CNN to classify images into multiple categories?

To build a CNN for multi-class image classification, I would create a neural network that extracts image features and predicts the correct category.

I would follow these steps:

Step 1: Collect the Dataset

Collect images belonging to different categories.

Example:

- Cat
- Dog
- Horse

Step 2: Preprocess the Images

- Resize images to a fixed size.
- Convert images into numerical arrays.

- Normalise pixel values.
- Divide the dataset into training, validation and test sets.

Step 3: Build the CNN Architecture

I would use:

- Convolutional Layers to extract features.
- ReLU Activation to introduce non-linearity.
- Max Pooling to reduce feature map dimensions.
- Flatten or Global Average Pooling to prepare features for classification.
- Dense Layers to perform classification.
- Softmax Activation in the output layer.

Step 4: Compile the Model

Choose:

- Optimiser: Adam.
- Loss Function: Categorical Cross-Entropy or Sparse Categorical Cross-Entropy.
- Evaluation Metric: Accuracy.

Step 5: Train the Model

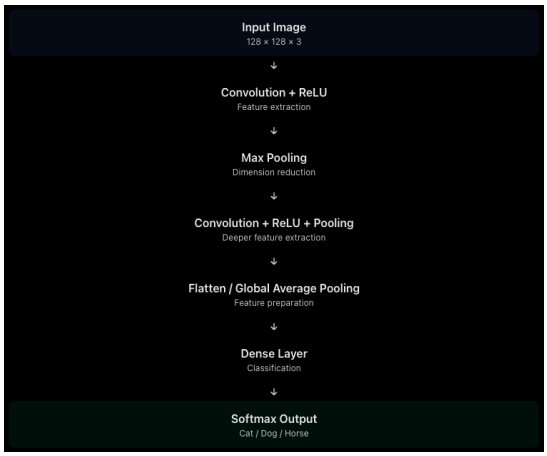
Train the CNN using the training dataset and monitor its validation performance.

Step 6: Evaluate the Model

Evaluate the trained model using unseen test images.

Check accuracy, precision, recall and the confusion matrix.

Example Architecture:



Key Point: A CNN extracts image features using convolutional layers and uses a classification layer with Softmax to predict the correct category.

61. What is LeNet and what was its contribution to computer vision?

LeNet is one of the earliest Convolutional Neural Network (CNN) architectures, developed by Yann LeCun and his research team.

It was mainly designed for handwritten digit recognition, such as recognising digits on cheques and postal codes.

Main layers of LeNet:

1. Convolutional layer – extracts image features.
2. Pooling layer – reduces the feature map size.
3. Convolutional layer – learns more complex features.
4. Pooling layer – reduces dimensions further.
5. Fully connected layers – perform classification.

Contribution to computer vision:

- Demonstrated that CNNs could automatically learn useful image features.
- Reduced the need for manually designed image features.
- Became a foundation for modern CNN architectures.

Key Point: LeNet was an important early CNN architecture that demonstrated the effectiveness of deep learning for image recognition.

62. What was the significance of AlexNet in Deep Learning?

AlexNet is a deep CNN architecture that achieved a major breakthrough in image classification during the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) in 2012.

It was developed by Alex Krizhevsky, Ilya Sutskever and Geoffrey Hinton.

Important features of AlexNet:

- Used multiple convolutional layers to extract image features.
- Used ReLU activation to improve training speed.
- Used dropout to reduce overfitting.
- Used max pooling to reduce feature map dimensions.

- Used GPU acceleration to train a large neural network.

Why was it significant?

AlexNet achieved a substantial improvement over previous image classification methods. It demonstrated that deep CNNs trained on large datasets could outperform traditional computer vision approaches.

Key Point: AlexNet popularised deep CNNs and helped establish deep learning as a major approach to computer vision.

63. What is the difference between VGG and Inception architectures?

VGG and Inception are CNN architectures used for image classification, but they have different approaches to extracting image features.

- VGG: Uses a simple and consistent architecture with multiple small convolution filters, mainly 3×3 filters.
- Inception: Uses multiple convolution operations with different filter sizes in parallel within the same module.

Example:

In VGG, convolutional layers are generally stacked one after another:

```
Input → Conv → Conv → Pooling → Conv → Conv → Pooling
```

In Inception, multiple operations happen in parallel:

```
Input → 1×1 Conv

Input → 3×3 Conv

Input → 5×5 Conv

Input → Pooling
```

The outputs are combined to create the final feature representation.

Feature	VGG	Inception
Architecture	Sequential	Parallel modules
Filters	Mainly 3×3	Multiple filter sizes
Parameters	Relatively high	More parameter-efficient
Complexity	Simple structure	More complex structure
Main idea	Deep layers of small filters	Capture features at different scales

Key Point: VGG focuses on stacking small convolutional layers, whereas Inception processes different filter sizes in parallel to capture features at multiple scales.

64. What is a residual connection, and how does ResNet use it?

A residual connection is a shortcut connection that allows the input of a layer or group of layers to bypass those layers and connect directly to a later layer.

ResNet (Residual Network) uses these connections to make very deep neural networks easier to train.

Problem with very deep networks:

As the number of layers increases, training can become difficult because of the vanishing gradient problem and degradation problem.

How does ResNet solve this?

Instead of learning the complete output directly, a residual block learns the difference between the desired output and the input.

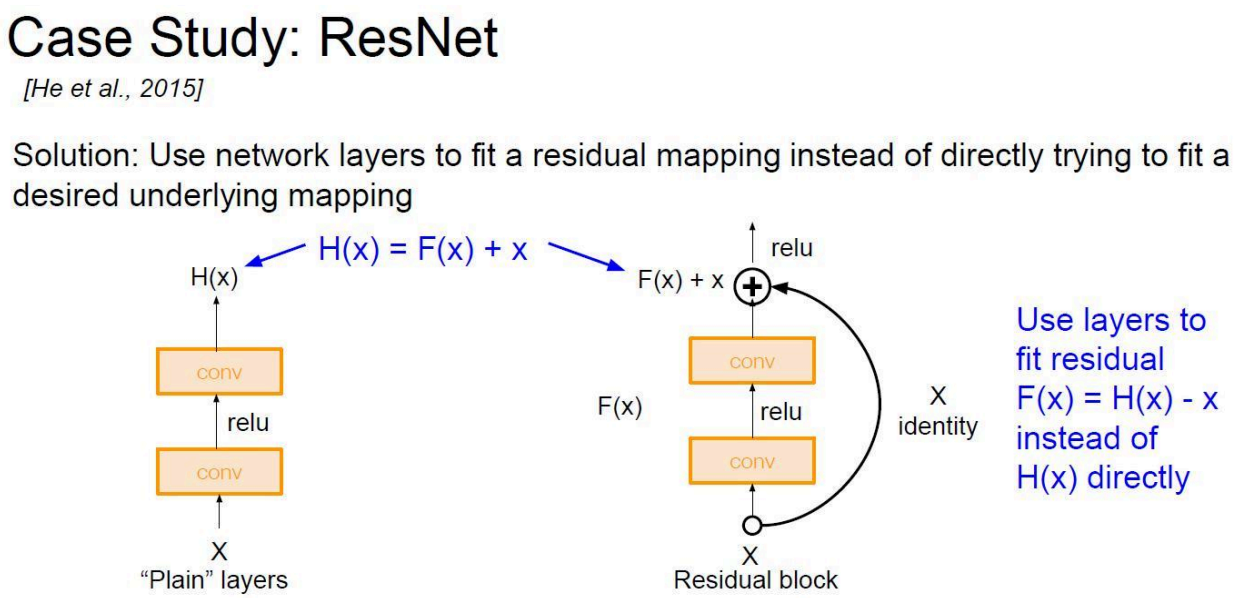
The mathematical representation is:

$$y = F(x) + x$$

Where:

- x = original input.
- $F(x)$ = output of the convolutional layers.
- y = final output after adding the shortcut connection.

Example:



The input travels through two paths:

1. Main path: passes through convolutional layers.
2. Shortcut path: bypasses those layers.

Both outputs are added together.

Key Point: Residual connections help gradients flow through deep networks and make it easier to train networks with many layers.

65. How does DenseNet differ from ResNet?

DenseNet and ResNet are both CNN architectures designed to improve information and gradient flow in deep networks.

However, they use different connection mechanisms.

ResNet:

ResNet uses shortcut connections that add the input to the output of a group of layers.

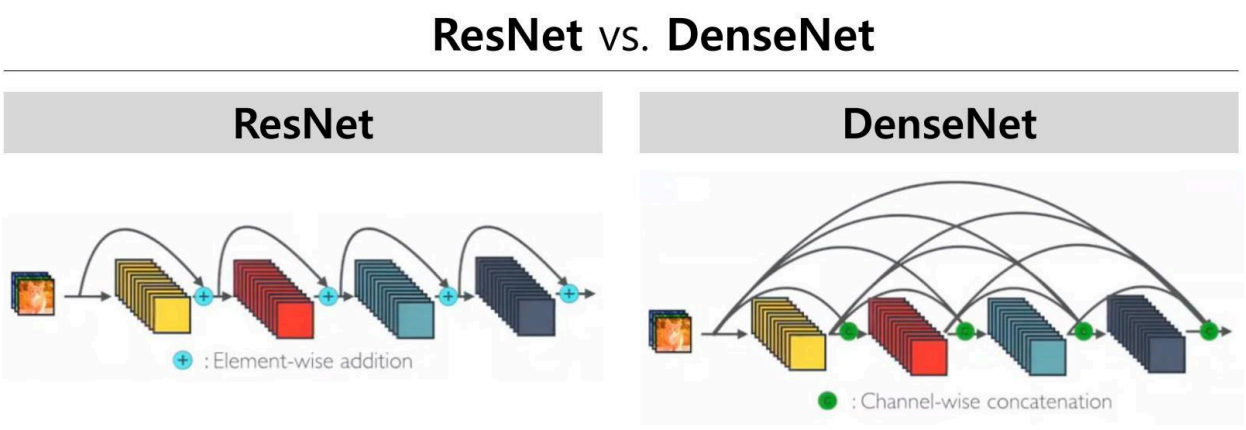
$$y = F(x) + x$$

DenseNet:

DenseNet connects each layer to every subsequent layer within a dense block. Each layer receives the feature maps from all previous layers through concatenation.

$$x_l = H_l([x_0, x_1, ..., x_{l-1}])$$

Here, [] represents concatenation.



Feature	ResNet	DenseNet
Connection	Shortcut connections	Dense connections

Operation	Addition	Concatenation
Information flow	Input added to output	Previous feature maps passed forward
Feature reuse	Good	Extensive
Memory usage	Generally lower	Can be higher due to concatenation

Key Point: ResNet uses addition to connect layers, whereas DenseNet uses concatenation to pass feature maps from earlier layers to later layers.

66. What is transfer learning, and why is it useful for image classification?

Transfer learning is a technique in which a model trained on one dataset is reused for a different but related task.

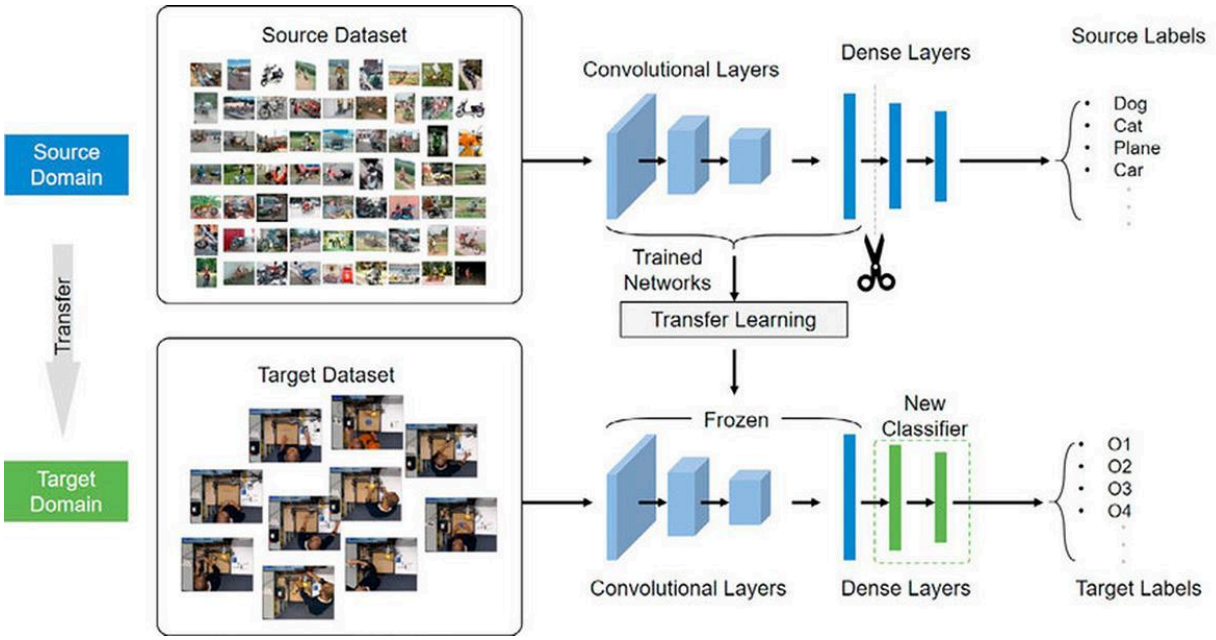
Instead of training a CNN from the beginning, we use a pre-trained model that has already learned useful image features.

For example, a model trained on ImageNet can recognise general features such as:

- Edges and lines.
- Shapes and textures.
- Patterns.
- Object parts.

We can reuse this knowledge to build a model that classifies different types of flowers, animals or vehicles.

How does it work?



1. Select a pre-trained model such as ResNet50, VGG16 or MobileNet.
2. Load its pre-trained weights.
3. Remove or replace the original classification layer.
4. Add new layers for the target classes.
5. Train the new classification layers.

Advantages:

- Requires less training data.
- Reduces training time.
- Often improves performance on small datasets.
- Reuses previously learned image features.

Key Point: Transfer learning allows us to reuse knowledge from a pre-trained model instead of training a CNN completely from scratch.

67. What is the difference between feature extraction and fine-tuning in transfer learning?

Feature extraction and fine-tuning are two common approaches to transfer learning.

1. Feature Extraction

In feature extraction, we use a pre-trained model as a fixed feature extractor.

- Load a pre-trained CNN.
- Freeze its existing convolutional layers.

- Replace the final classification layer.
- Train only the newly added layers.

The pre-trained weights remain unchanged.

2. Fine-Tuning

In fine-tuning, we allow some or all of the pre-trained layers to be trained again using the new dataset.

- Load a pre-trained CNN.
- Replace the classification layer.
- Unfreeze selected layers.
- Train them using the new dataset with a small learning rate.

Example:

Suppose we have a pre-trained ResNet50 model and want to classify cats and dogs.

- Feature extraction: Keep the ResNet50 feature layers frozen and train a new cats-and-dogs classification layer.
- Fine-tuning: Unfreeze some of the later ResNet50 layers and train them along with the new classification layer.

Feature	**Feature Extraction**	**Fine-Tuning**
Pre-trained layers	Frozen	Some or all are unfrozen
Weight updates	New layers only	Selected pre-trained layers and new layers
Training time	Usually shorter	Usually longer
Data requirement	Often works with smaller datasets	Needs careful regularisation

Main purpose	Reuse existing features	Adapt existing features

Key Point: Feature extraction keeps the pre-trained layers frozen, whereas fine-tuning updates some of their weights to adapt the model to the new task.

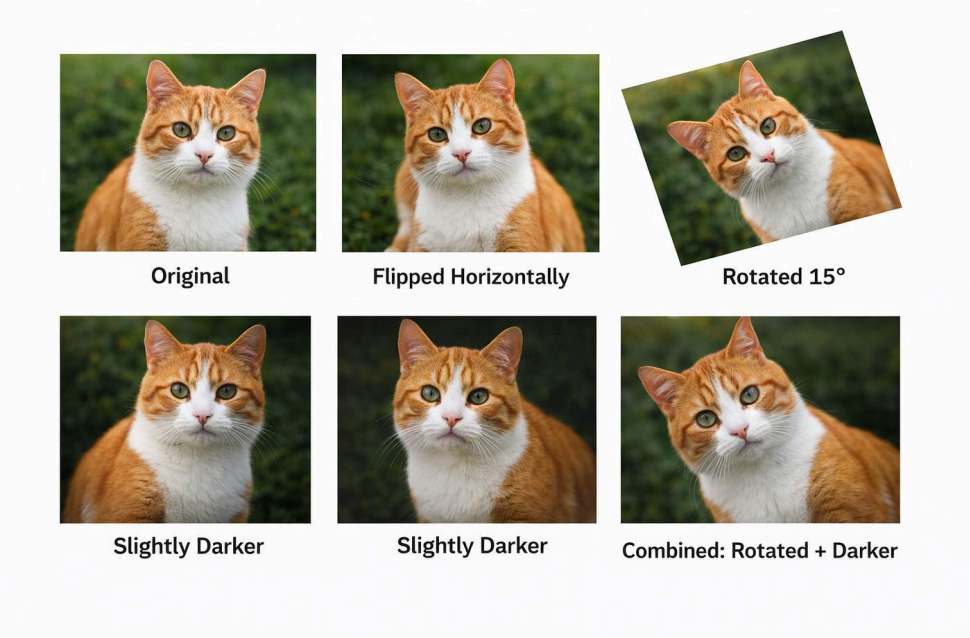
68. What is data augmentation, and which augmentation techniques are useful for images?

Data augmentation is a technique used to increase the diversity of a training dataset by creating modified versions of existing images.

Instead of collecting more images, we apply transformations to the available images.

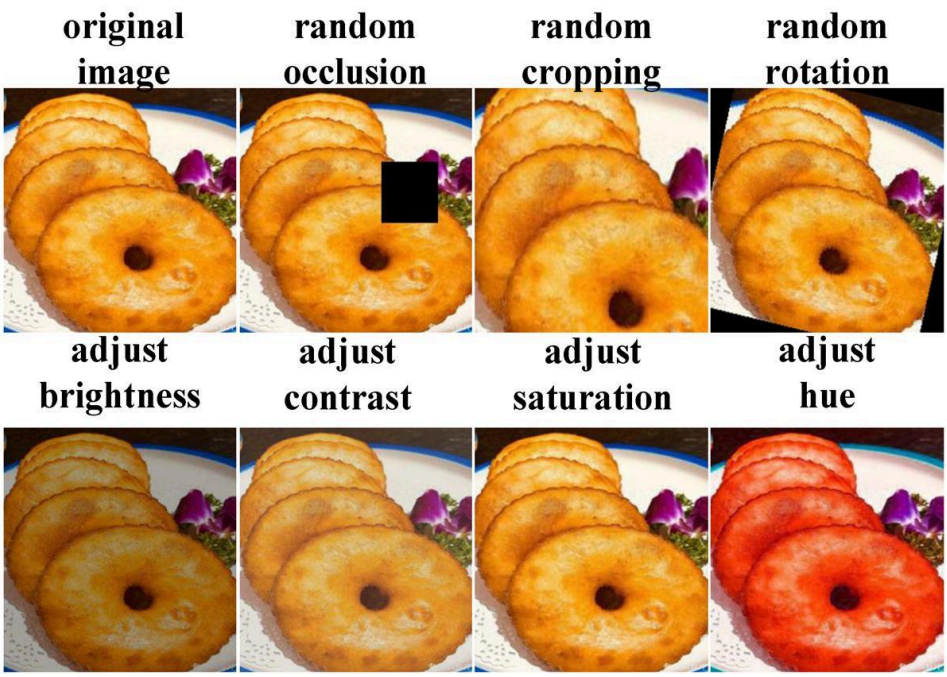
Common image augmentation techniques:

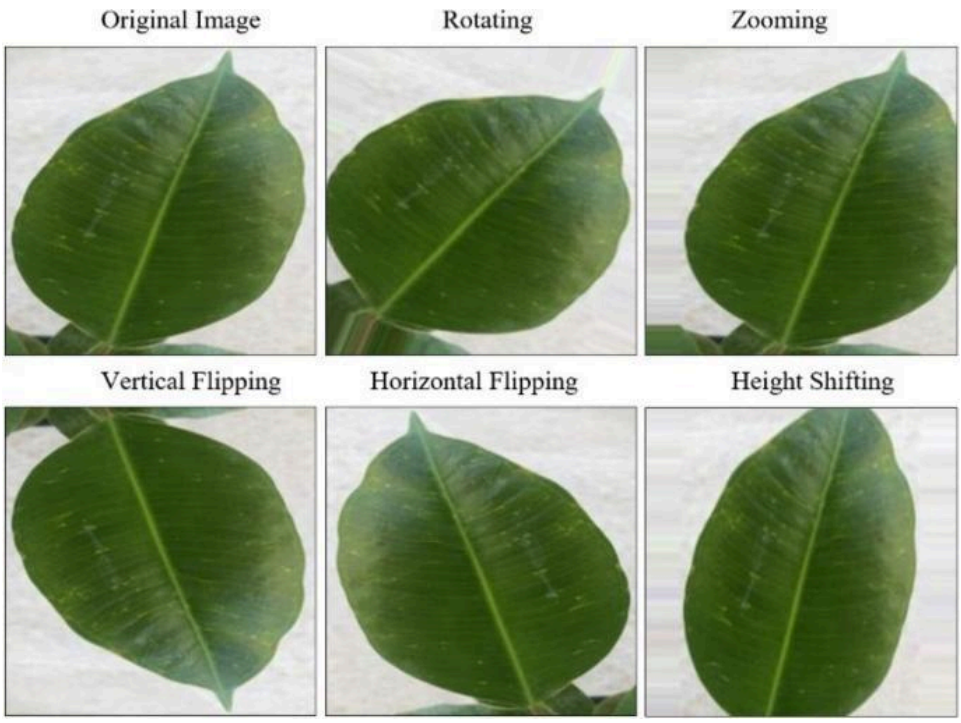
1. Rotation: Rotates an image by a certain angle.
2. Horizontal Flip: Flips an image from left to right.
3. Vertical Flip: Flips an image upside down.
4. Zoom: Enlarges or reduces the image.
5. Width and Height Shift: Moves the image horizontally or vertically.
6. Brightness Adjustment: Makes an image brighter or darker.
7. Random Cropping: Selects different portions of an image.
8. Shearing: Changes the shape of an image by shifting one side.



Original	Flip	Rotation	Random crop
			
• Image without any modification	• Flipped with respect to an axis for which the meaning of the image is preserved	• Rotation with a slight angle • Simulates incorrect horizon calibration	• Random focus on one part of the image • Several random crops can be done in a row

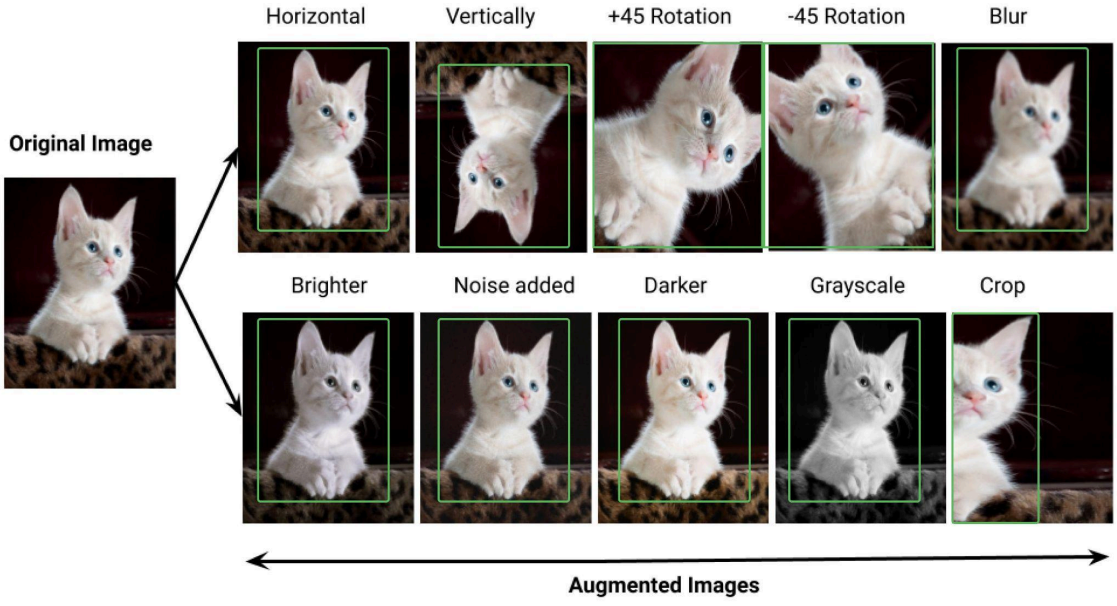
Color shift	Noise addition	Information loss	Contrast change
			
• Nuances of RGB is slightly changed • Captures noise that can occur with light exposure	• Addition of noise • More tolerance to quality variation of inputs	• Parts of image ignored • Mimics potential loss of parts of image	• Luminosity changes • Controls difference in exposition due to time of day

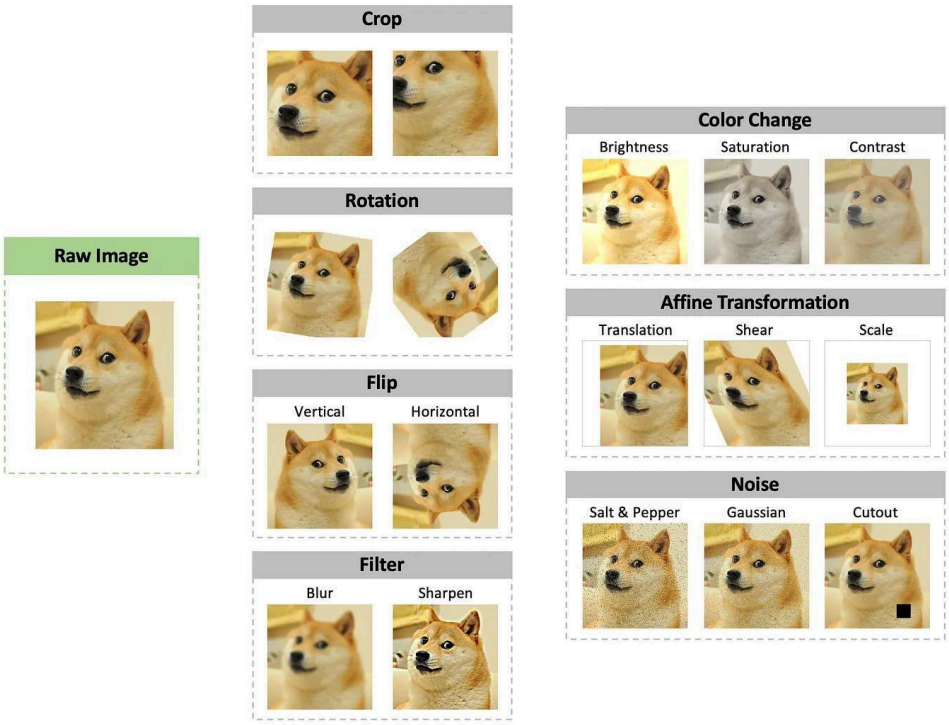



Original Image
Rotating
Zooming
Vertical Flipping
Horizontal Flipping
Height Shifting


Original Image
datagen_rotation
datagen_horizontal_flip
datagen_zoom
datagen_shift
datagen_brightness
datagen_shear
datagen_channel_shift


a. Original image
b. Rotation
c. Scale
d. Vertical Flip
e. Horizontal Flip
f. Diagonal Flip
g. Brightness adjustment
h. Crop
i. Add Gaussian noise


Original Image
Horizontal
Vertically
+45 Rotation
-45 Rotation
Blur
Brighter
Noise added
Darker
Grayscale
Crop
Augmented Images



8

Example:

Suppose we have 100 images of cats.

By applying rotation, flipping, zooming and brightness changes, we can generate additional training variations from those images.

Advantages:

- Helps reduce overfitting.
- Improves model generalisation.
- Increases training data diversity.
- Helps the model handle variations in real-world images.

Important: Augmentation should be applied to training data only. Validation and test images should normally remain unchanged, apart from standard preprocessing.

Also, choose transformations that preserve the image's correct label. For example, vertical flipping may not be suitable for every medical or object-recognition task.

Key Point: Data augmentation creates realistic variations of training images to help CNNs learn more robust features.

69. What is the difference between image classification, object detection and image segmentation?

These are three important computer vision tasks, but each has a different purpose.

1. Image Classification

Image classification identifies what an image contains.

Example: Given an image of a dog, the model predicts the label as `Dog`.

2. Object Detection

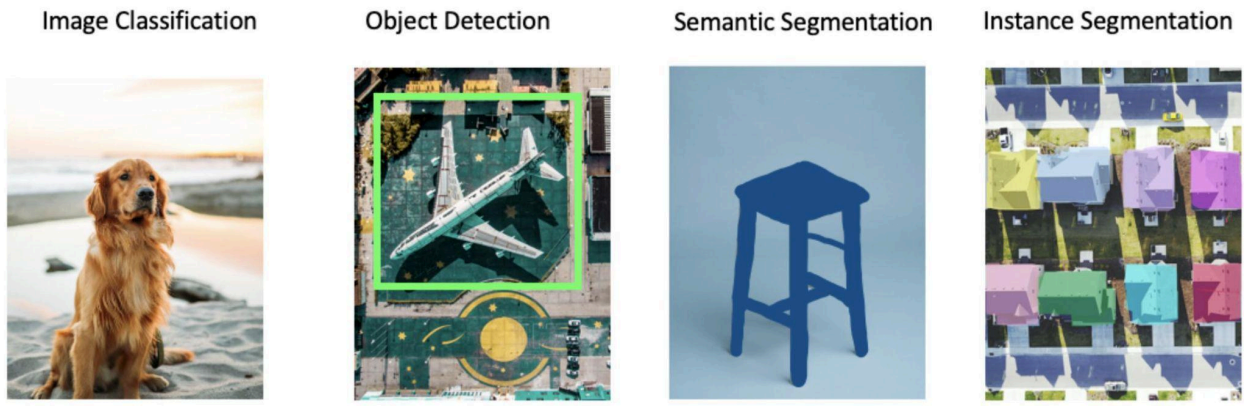
Object detection identifies objects and also locates them using bounding boxes.

Example: In an image containing two dogs and one cat, the model identifies each object and draws a bounding box around it.

3. Image Segmentation

Image segmentation classifies individual pixels in an image to identify the exact region occupied by an object.

Example: In an image of a dog, the model identifies the exact pixels belonging to the dog rather than drawing only a rectangular box.



Feature	Classification	Object Detection	Segmentation
Purpose	Identifies image category	Identifies and locates objects	Identifies exact object regions

Output	Class label	Class labels and bounding boxes	Pixel-level labels or masks
Location information	No	Yes	Yes, at pixel level
Example	Cat or dog	Detect cats and dogs	Identify exact cat and dog pixels

Real-world applications:

- Classification: Identifying whether an X-ray image shows a particular condition.
- Object detection: Detecting vehicles and pedestrians in autonomous driving.
- Segmentation: Identifying roads, buildings and vehicles at pixel level in self-driving cars.

Key Point: Classification tells us what is in an image, detection tells us what and where, and segmentation identifies the exact pixels belonging to objects or regions.

70. How would you handle a small, imbalanced image dataset when training a CNN?

A small, imbalanced image dataset means that we have limited training images and some classes contain significantly more images than others.

For example:

- Cat: 1,000 images.
- Dog: 200 images.
- Horse: 50 images.

In this situation, the model may become biased towards the majority class and perform poorly on minority classes.

Steps to handle this problem:

1. Data Augmentation

Apply rotation, flipping, zooming and brightness adjustments to increase the diversity of training images.

2. Transfer Learning

Use a pre-trained model such as ResNet50, MobileNet or EfficientNet instead of training a CNN from scratch.

3. Class Weights

Assign higher weights to minority classes during training so that mistakes on those classes have a greater impact on the loss.

Example using Keras:

```
None
class_weights = {
    0: 1.0,
    1: 3.0,
    2: 5.0
}

model.fit(
    train_data,
    validation_data=val_data,
    epochs=20,
    class_weight=class_weights
)
```

4. Oversampling

Increase the representation of minority classes by sampling more of their training images. Apply augmentation to create additional variations rather than simply duplicating the same images.

5. Proper Evaluation Metrics

Do not rely only on accuracy. Use:

- Precision
- Recall
- F1-score
- Confusion matrix
- Per-class performance

6. Proper Dataset Splitting

Split the dataset into training, validation and test sets using stratification to maintain class proportions. Keep images from the same original source or subject within one split to avoid data leakage.

7. Fine-Tuning

After training the new classification layers, unfreeze selected layers of the pre-trained model and fine-tune them using a low learning rate.

Key Point: For a small and imbalanced image dataset, I would use transfer learning, data augmentation, class weights, stratified splitting and suitable evaluation metrics to improve model performance.

71. What is a Recurrent Neural Network (RNN), and what types of problems is it designed to solve?

A Recurrent Neural Network (RNN) is a type of neural network designed to process sequential data, where the order of information is important.

Unlike a traditional neural network, an RNN can use information from previous time steps to understand the current input.

Examples of sequential data:

- Text and sentences.
- Speech and audio.
- Stock market prices.
- Weather measurements.
- Time-series data.

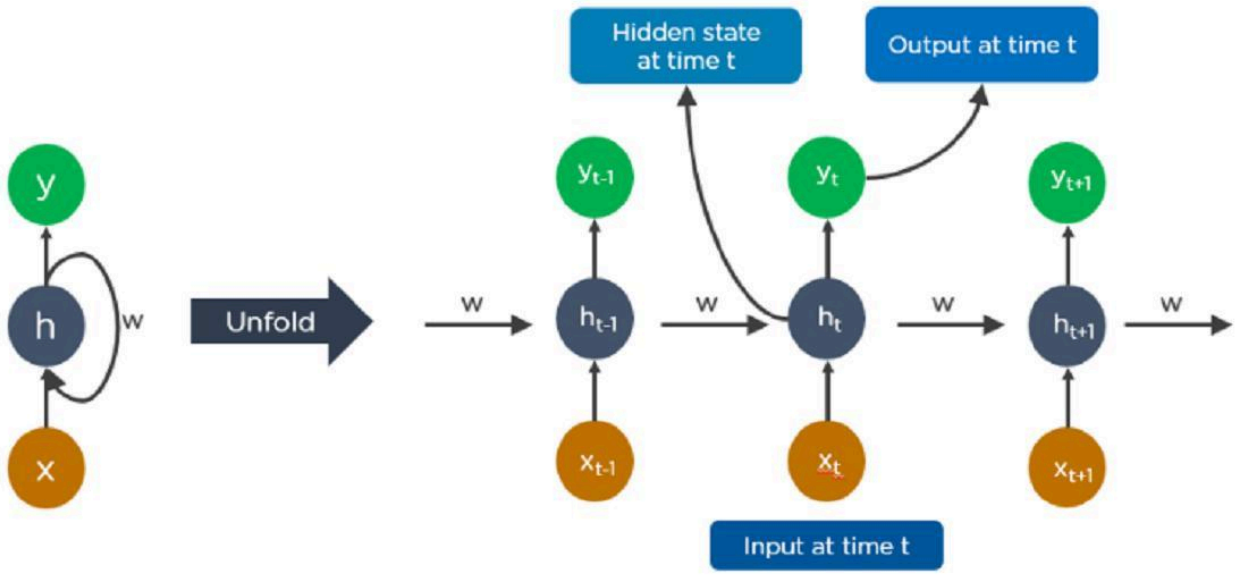
Real-world example:

Consider the sentence:

"The weather is very _____."

To predict the next word, an RNN processes the previous words and uses their information to predict a suitable word, such as "good".

How an RNN works:



1. Receives an input at a particular time step.
2. Processes the input using its current hidden state.
3. Passes the updated hidden state to the next time step.
4. Repeats this process for the remaining sequence.
5. Produces an output based on the processed information.

Key Point: RNNs are designed to process sequential data by maintaining information from previous time steps.

72. How does an RNN maintain information from previous time steps?

An RNN maintains information from previous time steps using a mechanism called the hidden state.

At each time step, the network receives two things:

1. Current input.
2. Hidden state from the previous time step.

It combines them to calculate a new hidden state, which is passed to the next time step.

Example:

Consider the sentence:

"I love machine learning."

The RNN processes the sentence word by word:

- Time step 1: "I"
- Time step 2: "love"
- Time step 3: "machine"
- Time step 4: "learning"

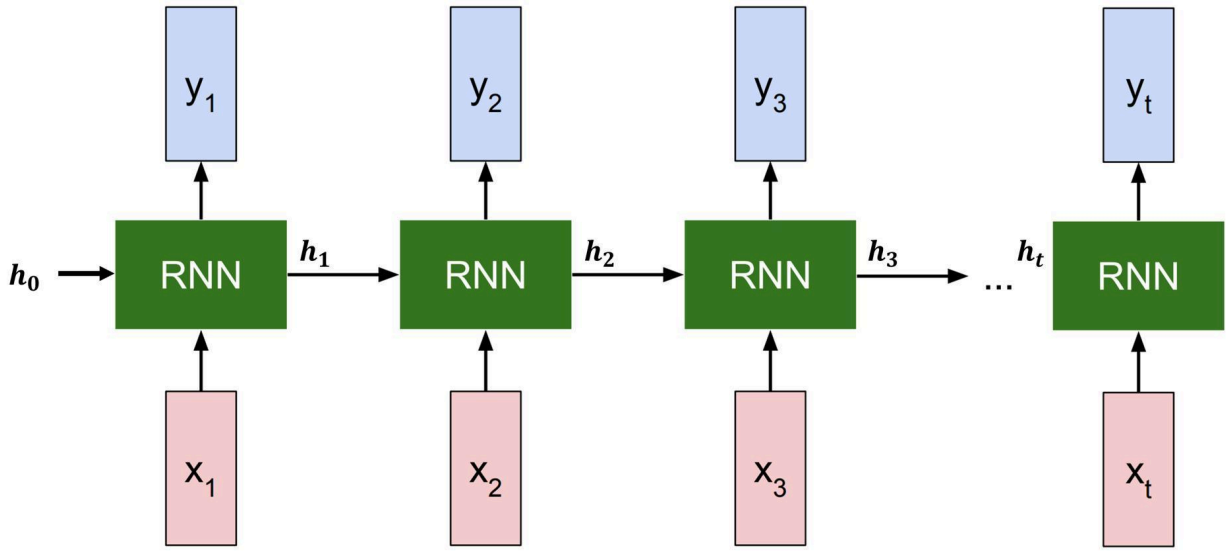
At each step, the RNN updates its hidden state using the current word and previous information.

Mathematical representation:

$$h_t = f(W_x x_t + W_h h_{t-1} + b)$$

Where:

- h_t = current hidden state.
- x_t = current input.
- h_{t-1} = previous hidden state.
- W_x, W_h = weight matrices.
- b = bias.
- f = activation function, commonly tanh.



Key Point: An RNN passes its hidden state from one time step to the next, allowing it to use previous information when processing new inputs.

73. What is a hidden state in an RNN?

A hidden state is an internal representation that stores information about the sequence processed by an RNN up to the current time step.

It acts as the network's memory.

At every time step, the hidden state is updated using the current input and the previous hidden state.

Example:

Consider the sentence:

"The student is studying Data Science."

When processing this sentence:

- After "The", the hidden state contains information about the first word.
- After "student", the hidden state is updated with information from both words.
- After "is studying", the hidden state contains information from the sequence processed so far.
- After "Data Science", it contains an updated representation of the complete sequence.

Important points:

- It carries information between time steps.
- It is updated whenever a new input is processed.
- It helps the network understand sequence context.
- It is used to calculate the output.

Key Point: The hidden state is the internal memory of an RNN that carries information from previous time steps.

74. What is Backpropagation Through Time (BPTT)?

Backpropagation Through Time (BPTT) is the training algorithm used to update the weights of a Recurrent Neural Network.

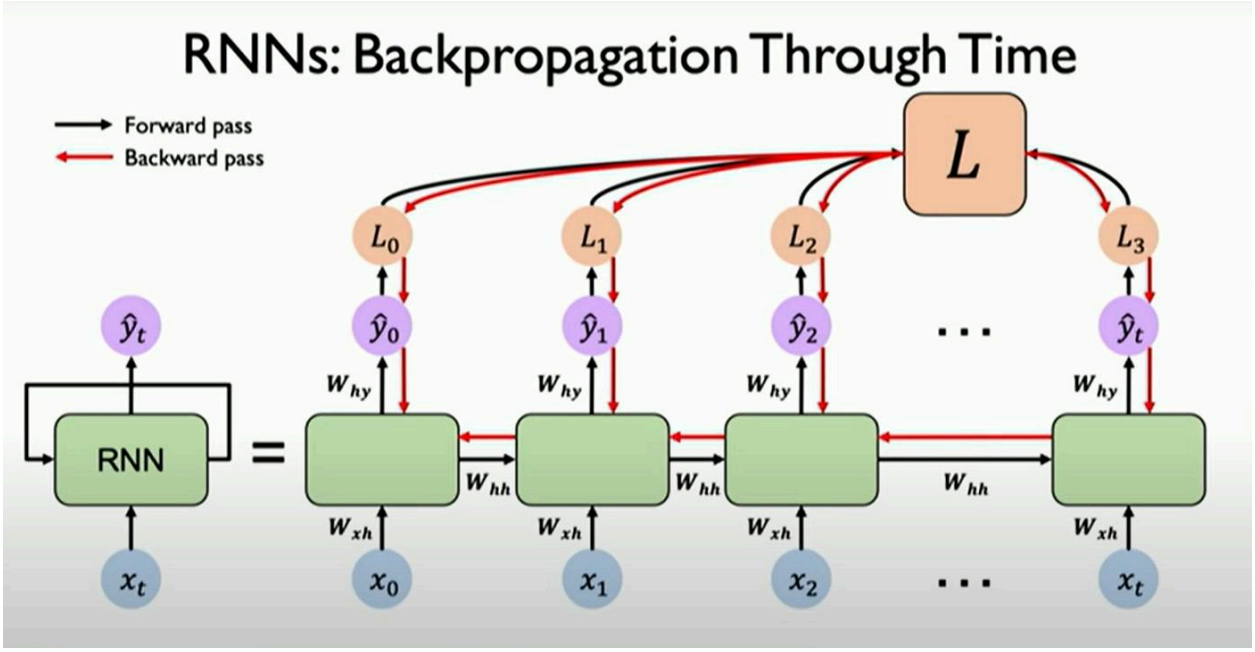
It is an extension of the traditional backpropagation algorithm, specifically designed for sequential networks.

In a standard neural network, backpropagation calculates gradients through layers. In an RNN, BPTT calculates gradients across both layers and time steps.

How does BPTT work?

1. The RNN processes the input sequence from the first time step to the last.

2. It generates predictions.
3. The loss is calculated by comparing predictions with actual outputs.
4. The network is conceptually unfolded across time steps.
5. The error is propagated backwards through the time steps.
6. Gradients are calculated and weights are updated.



Example:

Suppose an RNN predicts the next word in a sentence.

If the predicted word is incorrect, BPTT helps determine how the weights across the previous time steps contributed to that error.

Key Point: BPTT trains RNNs by propagating errors backwards through the sequence of time steps and updating the network weights.

75. Why do standard RNNs struggle with long-term dependencies?

Standard RNNs struggle with long-term dependencies because they have difficulty retaining information from much earlier time steps when processing long sequences.

This happens mainly because of the vanishing gradient problem.

Example:

Consider the sentence:

"I grew up in India. I speak fluent Hindi."

To understand why the person speaks Hindi, the model may need to remember information from the earlier part of the sentence.

However, a standard RNN may struggle to retain that information across many time steps.

Main reasons:

1. Vanishing Gradients: Gradients become very small during backpropagation, making it difficult to learn relationships between distant time steps.
2. Repeated Processing: Information passes through the recurrent network repeatedly.
3. Limited Memory: Standard RNNs do not have a dedicated memory mechanism for selectively retaining important information.

How can this problem be addressed?

- LSTM networks.
- GRU networks.
- Suitable gradient management techniques.

Key Point: Standard RNNs struggle with long-term dependencies because information and gradients can become weaker as they pass through many time steps.

76. How do vanishing and exploding gradients affect RNN training?

Vanishing and exploding gradients are two major problems that can occur when training deep neural networks and RNNs.

They affect how the network learns by changing the size of gradients during backpropagation.

1. Vanishing Gradient Problem

The vanishing gradient problem occurs when gradients become extremely small during backpropagation.

As a result:

- Earlier layers receive very small updates.
- The network learns very slowly.
- Long-term dependencies become difficult to learn.

Example:

Suppose an RNN processes a sequence of 100 words. When the error is propagated backwards, the gradients may become extremely small by the time they reach the first few words.

The network may struggle to learn relationships between the beginning and end of the sentence.

2. Exploding Gradient Problem

The exploding gradient problem occurs when gradients become extremely large during backpropagation.

As a result:

- Weight updates become excessively large.
- Training becomes unstable.
- The loss may increase or become `NaN`.

How can we solve these problems?

Problem	Possible solution
Vanishing gradients	LSTM and GRU architectures, suitable activation and initialisation
Exploding gradients	Gradient clipping, suitable learning rate

Gradient clipping example:

```
None
import tensorflow as tf

optimizer = tf.keras.optimizers.Adam(
    learning_rate=0.001,
    clipnorm=1.0
)
```

Here, `clipnorm=1.0` limits the gradient norm to help prevent excessively large updates.

Key Point: Vanishing gradients make learning difficult because gradients become too small, while exploding gradients make training unstable because gradients become too large.

77. What is a Long Short-Term Memory (LSTM) network?

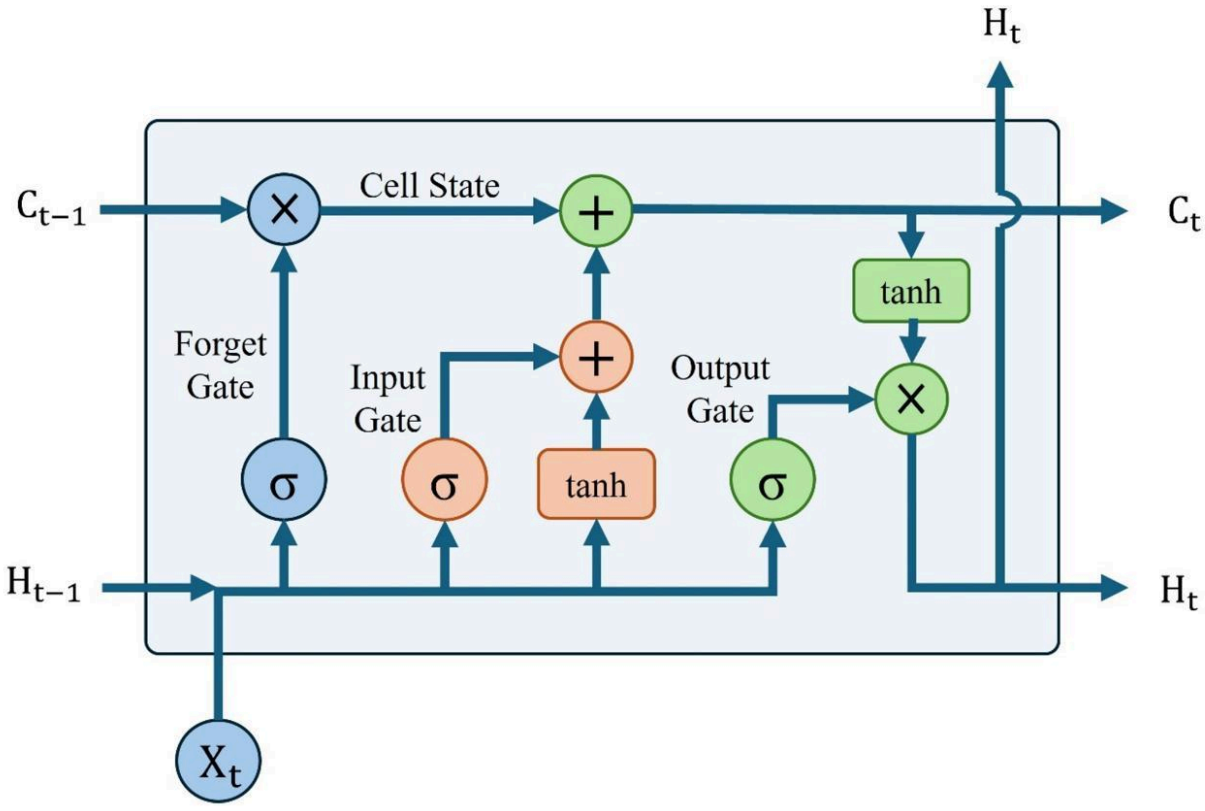
Long Short-Term Memory (LSTM) is a special type of Recurrent Neural Network designed to learn long-term dependencies in sequential data.

It was introduced to address the limitations of standard RNNs, particularly the vanishing gradient problem.

Unlike a standard RNN, LSTM uses a dedicated memory mechanism called the cell state and three gates to control information flow.

Main components of LSTM:

1. Cell State: Carries long-term information through the sequence.
2. Forget Gate: Decides which previous information should be removed.
3. Input Gate: Decides which new information should be stored.
4. Output Gate: Decides which information should be passed to the next hidden state.



Example:

Suppose we want to predict the next word in a long paragraph.

An LSTM can retain important information from earlier sentences and use it when processing later sentences.

Applications:

- Text generation.
- Language modelling.
- Speech recognition.
- Time-series forecasting.
- Sentiment analysis.

Key Point: LSTM is an improved RNN architecture that uses cell states and gates to retain important information over longer sequences.

78. What are the input, forget and output gates in an LSTM?

LSTM uses three main gates to control the flow of information through its memory.

Each gate uses a sigmoid activation function to produce values between 0 and 1.

A value closer to 0 means less information passes through, while a value closer to 1 means more information passes through.

1. Forget Gate

The forget gate decides which information from the previous cell state should be retained or removed.

Example:

If an old piece of information is no longer useful for predicting the next word, the forget gate can reduce its importance.

2. Input Gate

The input gate decides which new information should be stored in the cell state.

Example:

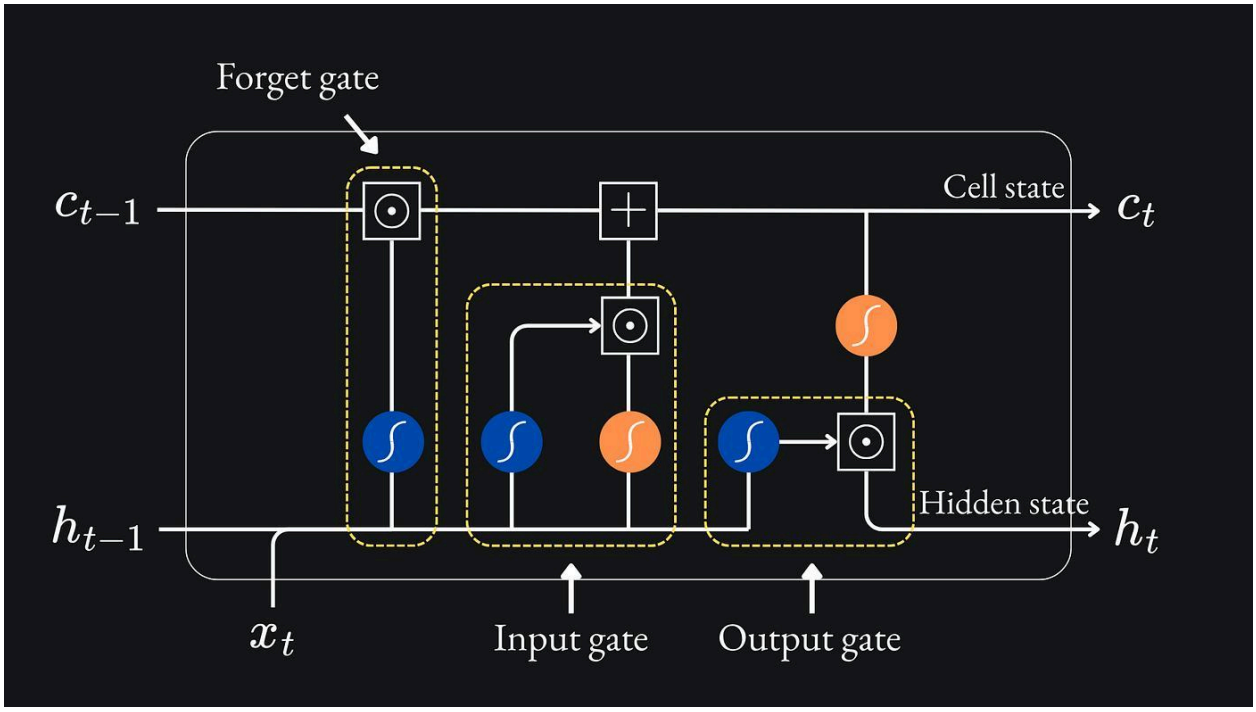
When processing a new word, the input gate determines which parts of that word's information are useful to remember.

3. Output Gate

The output gate decides which information from the cell state should be used to calculate the current hidden state.

Example:

It determines which stored information should be used to produce the next prediction.



Gate	Main function
Forget Gate	Removes or retains previous information
Input Gate	Selects new information to store
Output Gate	Selects information for the current output

Key Point: The forget gate controls old information, the input gate controls new information, and the output gate controls what information is used as the current output.

79. What is a Gated Recurrent Unit (GRU), and how does it differ from LSTM?

A Gated Recurrent Unit (GRU) is a type of Recurrent Neural Network designed to capture long-term dependencies in sequential data.

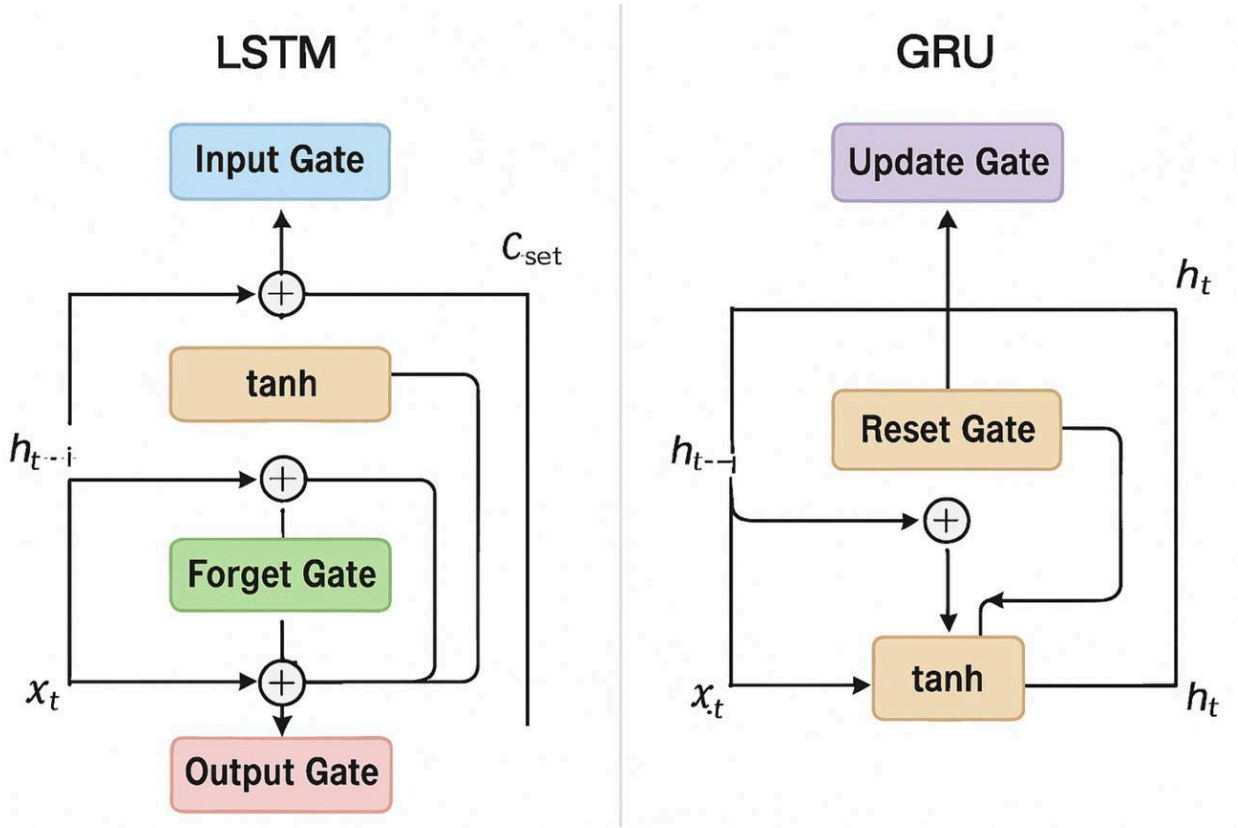
Like LSTM, GRU uses gates to control information flow and help reduce the vanishing gradient problem.

However, GRU has a simpler architecture than LSTM.

GRU has two main gates:

1. Update Gate: Decides how much previous information should be retained and how much new information should be incorporated.
2. Reset Gate: Controls how much previous information should be ignored when calculating new candidate information.

Unlike LSTM, GRU does not maintain a separate cell state. It uses the hidden state to carry information.



Difference between LSTM and GRU:

Feature	LSTM	GRU
Gates	Three	Two
Gates used	Forget, input, output	Update, reset
Cell state	Separate cell state	No separate cell state
Architecture	More complex	Simpler
Parameters	Generally more	Generally fewer
Training speed	Can be slower	Often faster

Example:

Suppose we are building a model to predict tomorrow's temperature using historical weather data.

Both LSTM and GRU can learn patterns from previous measurements. GRU may be a useful choice when we want a simpler model with fewer parameters.

Key Point: GRU is a simplified version of gated recurrent architecture that uses two gates and does not maintain a separate cell state.

80. When would you choose an RNN, LSTM or GRU for a sequence prediction task?

The choice between RNN, LSTM and GRU depends on the sequence length, complexity of the task, available training data and computational resources.

1. Choose a Standard RNN when:

- The sequence is relatively short.
- The task has simple sequential patterns.
- Long-term dependencies are not important.
- You need a basic model to understand sequential processing.

Example: Predicting a value based on a short sequence of measurements.

2. Choose LSTM when:

- The sequence is long.
- Long-term dependencies are important.
- Earlier information needs to be retained for later predictions.
- The task requires more complex memory management.

Example: Language modelling or long-term time-series forecasting.

3. Choose GRU when:

- You need to capture sequential dependencies.
- You want a simpler architecture than LSTM.
- Training efficiency is important.
- You want to reduce the number of model parameters.

Example: Short- or medium-length time-series forecasting.

Comparison:

Feature	**RNN**	**LSTM**	**GRU**
Architecture	Simple	Complex	Moderate
Memory mechanism	Hidden state	Hidden and cell states	Hidden state
Long-term dependencies	Limited	Good	Good

Gates	None	Three	Two
Computational cost	Usually lower	Usually higher	Often lower than LSTM
Best suited for	Simple sequences	Complex, longer dependencies	Efficient sequence learning

Practical recommendation:

If I were developing a sequence prediction model, I would:

1. Start with a simple baseline.
2. Try a GRU for a relatively efficient gated model.
3. Try an LSTM if retaining longer-term information is important.
4. Compare their validation performance, training time and computational requirements.

Key Point: Standard RNNs are suitable for simpler sequential tasks, LSTMs are useful for learning longer-term dependencies, and GRUs offer a simpler gated alternative.

81. How is Deep Learning used in Natural Language Processing (NLP)?

Deep Learning is used in Natural Language Processing (NLP) to help computers understand, process and generate human language.

Traditional NLP systems often depend on manually designed features and rules. Deep Learning models can automatically learn useful language patterns from large amounts of text data.

Common applications of Deep Learning in NLP:

1. Sentiment Analysis: Identifies whether a sentence expresses positive, negative or neutral sentiment.
2. Machine Translation: Translates text from one language to another.
3. Text Classification: Categorises documents, emails or messages.
4. Text Generation: Generates human-like text.
5. Named Entity Recognition (NER): Identifies names, locations, organisations and other entities.
6. Question Answering: Answers questions using information from text.

7. Text Summarisation: Generates shorter versions of longer documents.
8. Speech Recognition: Converts spoken language into text.

Common Deep Learning architectures used in NLP:

- RNN
- LSTM
- GRU
- CNN
- Transformer

Example:

Consider a customer review:

"The product quality is excellent, and I am very happy."

A Deep Learning sentiment classification model can analyse this sentence and predict:

```
Sentiment: Positive
```

Key Point: Deep Learning helps NLP systems learn language patterns automatically and perform tasks such as sentiment analysis, translation, text generation and question answering.

82. What is tokenisation, and why is it important for neural language models?

Tokenisation is the process of breaking text into smaller units called tokens.

A token can be a complete word, part of a word, a character or even a punctuation mark, depending on the tokenisation method.

Neural language models cannot directly process raw text. Tokenisation converts text into units that can be mapped to numerical IDs and processed by the model.

Example:

Original sentence:

```
I love Deep Learning.
```

Word-level tokenisation:

```
None
["I", "love", "Deep", "Learning", "."]
```

The tokens are then converted into numerical IDs using a vocabulary.

Example:

```
None
{
    "I": 1,
    "love": 2,
    "Deep": 3,
    "Learning": 4,
    ".": 5
}
```

The resulting sequence is:

```
None
[1, 2, 3, 4, 5]
```

Common types of tokenisation:

1. Word-level: Splits text into individual words.
2. Character-level: Splits text into individual characters.
3. Subword-level: Splits words into smaller meaningful units.

Modern Transformer models commonly use subword tokenisation methods such as BPE (Byte Pair Encoding) and WordPiece.

Why is tokenisation important?

- Converts text into manageable units.
- Helps models process unknown or complex words.
- Creates numerical representations of text.
- Provides the input sequence required by neural language models.

Key Point: Tokenisation converts raw text into tokens, which can then be converted into numerical IDs for neural language models.

83. What is a word embedding?

A word embedding is a numerical representation of a word in the form of a dense vector.

Instead of representing each word using a separate binary value, word embeddings represent words using a set of continuous numerical values.

The main purpose is to capture semantic and contextual relationships between words.

Example:

Consider these words:

- King
- Queen
- Man
- Woman

A word embedding model converts each word into a vector.

For example:

```
None
King  = [0.25, 0.81, 0.42, 0.63]
Queen = [0.27, 0.78, 0.45, 0.67]
Man   = [0.18, 0.72, 0.31, 0.52]
Woman = [0.20, 0.75, 0.35, 0.56]
```

These are illustrative vectors, not actual trained embeddings.

Words with similar meanings or usage patterns tend to have similar vector representations.

Advantages of word embeddings:

- Capture semantic relationships.
- Represent words using fewer dimensions than one-hot encoding.
- Help neural networks understand similarities between words.
- Improve performance in many NLP tasks.

Common word embedding techniques:

- Word2Vec
- GloVe
- FastText

- Embedding layers in neural networks.

Key Point: Word embeddings convert words into dense numerical vectors that help neural networks learn relationships between words.

84. What is the difference between one-hot encoding and dense word embeddings?

One-hot encoding and dense word embeddings are two methods used to represent words numerically.

However, they differ in their representation, dimensionality and ability to capture semantic relationships.

1. One-Hot Encoding

One-hot encoding represents each word using a binary vector.

Each word receives a vector with one `1` and all other values as `0`.

Example:

Suppose the vocabulary contains four words:

```
["Apple", "Mango", "Orange", "Banana"]
```

Their representations are:

```
None
Apple  = [1, 0, 0, 0]
Mango  = [0, 1, 0, 0]
Orange = [0, 0, 1, 0]
Banana = [0, 0, 0, 1]
```

Each word has a separate binary representation.

2. Dense Word Embeddings

Dense embeddings represent words using smaller vectors containing continuous numerical values.

Example:

```
None
Apple  = [0.25, 0.61, 0.32]
Mango  = [0.28, 0.58, 0.35]
Orange = [0.22, 0.64, 0.30]
Banana = [0.31, 0.55, 0.38]
```

These are illustrative values.

Words with similar meanings or usage patterns can have similar vector representations.

Comparison:

Feature	One-Hot Encoding	Dense Word Embeddings
Representation	Binary vectors	Continuous numerical vectors
Dimensions	Vocabulary size	Usually much smaller
Values	0 and 1	Real-valued numbers
Semantic relationships	Not captured directly	Can be learned
Sparsity	Sparse	Dense
Storage	Can be expensive for large vocabularies	More compact

Key Point: One-hot encoding represents words as separate binary vectors, whereas dense embeddings represent words as compact vectors that can capture semantic relationships.

85. How do Word2Vec and GloVe generate word representations?

Word2Vec and GloVe are popular techniques used to generate dense word embeddings.

Both methods learn numerical representations of words based on their relationships with other words in a text corpus.

However, they use different learning approaches.

1. Word2Vec

Word2Vec is a neural network-based technique that learns word embeddings by predicting relationships between words in a sentence.

It has two main architectures:

- CBOW (Continuous Bag of Words): Predicts a target word using its surrounding context words.
- Skip-gram: Predicts surrounding context words using a target word.

Example:

Sentence:

```
I love machine learning
```

In Skip-gram, if the target word is `machine`, the model tries to predict nearby words such as `love` and `learning`.

2. GloVe (Global Vectors)

GloVe is a word embedding technique that learns word representations using global word co-occurrence statistics.

It analyses how frequently words appear together across a large text corpus.

Example:

If the words `machine` and `learning` frequently appear together, GloVe uses that co-occurrence information to learn their vector representations.

Comparison:

Feature	Word2Vec	GloVe
Approach	Predictive	Count-based with an optimisation objective
Learning method	Predicts context relationships	Uses global co-occurrence statistics
Main architectures	CBOW and Skip-gram	Co-occurrence matrix-based
Context	Local context windows	Global corpus statistics
Output	Dense word vectors	Dense word vectors

Key Point: Word2Vec learns embeddings by predicting words from their context or vice versa, while GloVe learns embeddings using global word co-occurrence statistics.

86. What is the difference between CBOW and Skip-gram in Word2Vec?

CBOW and Skip-gram are two architectures used by Word2Vec to learn word embeddings.

Their main difference is the direction of prediction.

1. CBOW (Continuous Bag of Words)

CBOW predicts the target word using the surrounding context words.

Example:

Sentence:

```
I love machine learning
```

Suppose the target word is `machine`.

The model receives surrounding context words:

```
love, learning
```

It tries to predict:

```
machine
```

2. Skip-gram

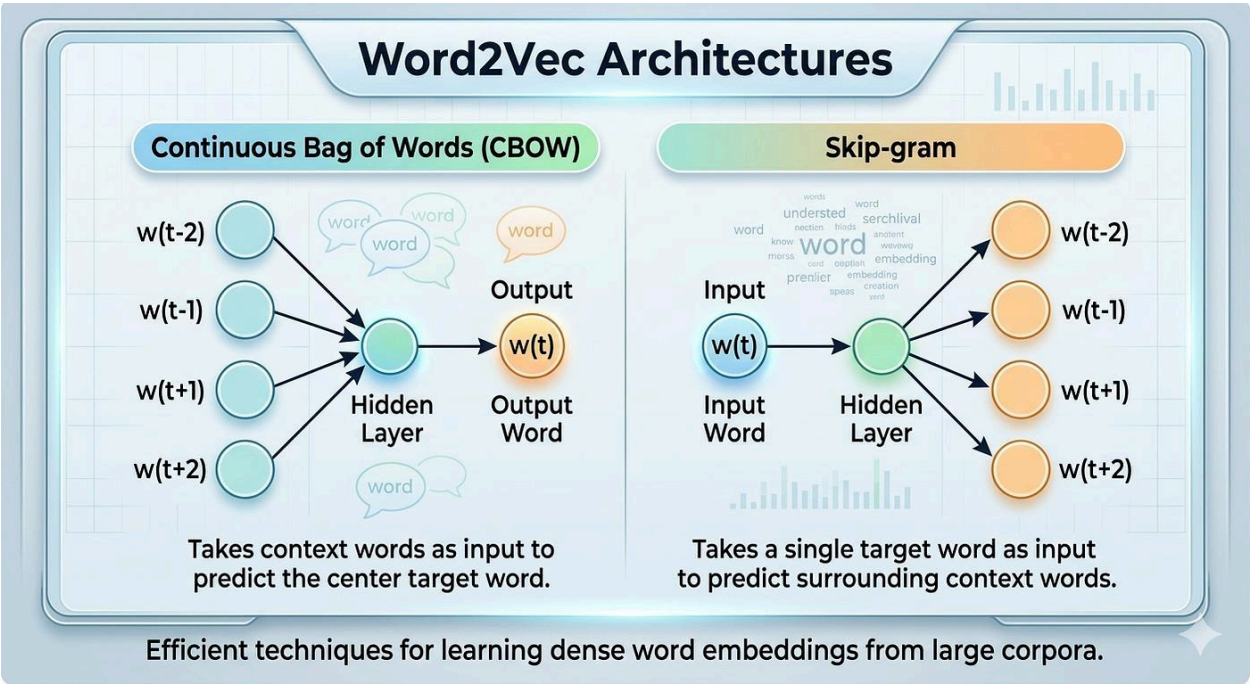
Skip-gram does the opposite. It uses the target word to predict the surrounding context words.

Using the same sentence:

```
I love machine learning
```

If the target word is `machine`, the model tries to predict nearby words such as:

```
love, learning
```



Comparison:

Feature	CBOW	Skip-gram
Input	Context words	Target word
Output	Target word	Context words
Prediction	Context → Target	Target → Context
Training	Generally faster	Generally slower
Data suitability	Often effective for frequent words	Often useful for rare words

Key Point: CBOW predicts a word from its surrounding context, whereas Skip-gram predicts the surrounding context from a given word.

87. What are the limitations of static word embeddings?

Static word embeddings assign a single vector representation to each word, regardless of how the word is used in different sentences.

Popular techniques such as Word2Vec and GloVe generate static word embeddings.

Their main limitation is that they cannot properly represent different meanings of the same word based on its context.

Example:

Consider the word `Bank`.

Sentence 1:

"I deposited money in the bank."

Here, `Bank` refers to a financial institution.

Sentence 2:

"We sat near the river bank."

Here, `Bank` refers to the land beside a river.

A static embedding model generally assigns the same vector to `Bank` in both sentences.

This makes it difficult for the model to understand the different meanings.

Other limitations:

1. No Contextual Understanding: The same word receives the same representation in different contexts.
2. Polysemy Problem: Cannot properly distinguish multiple meanings of a word.
3. Out-of-Vocabulary Words: Traditional static embedding models may struggle with words that were not present in their training vocabulary.
4. Limited Contextual Information: They do not dynamically adjust word representations based on surrounding words.

How can this be improved?

Contextual embeddings are used in modern language models such as:

- BERT.
- GPT.
- ELMo.

These models generate word or token representations based on the surrounding context.

Key Point: Static embeddings assign one fixed vector to each word, whereas contextual embeddings generate representations based on how the word is used in a sentence.

88. What is an encoder–decoder architecture, and how is it used in sequence-to-sequence tasks?

An encoder–decoder architecture is a neural network structure used to convert one sequence into another sequence.

It is commonly used in sequence-to-sequence (Seq2Seq) tasks, where the input and output are both sequential data.

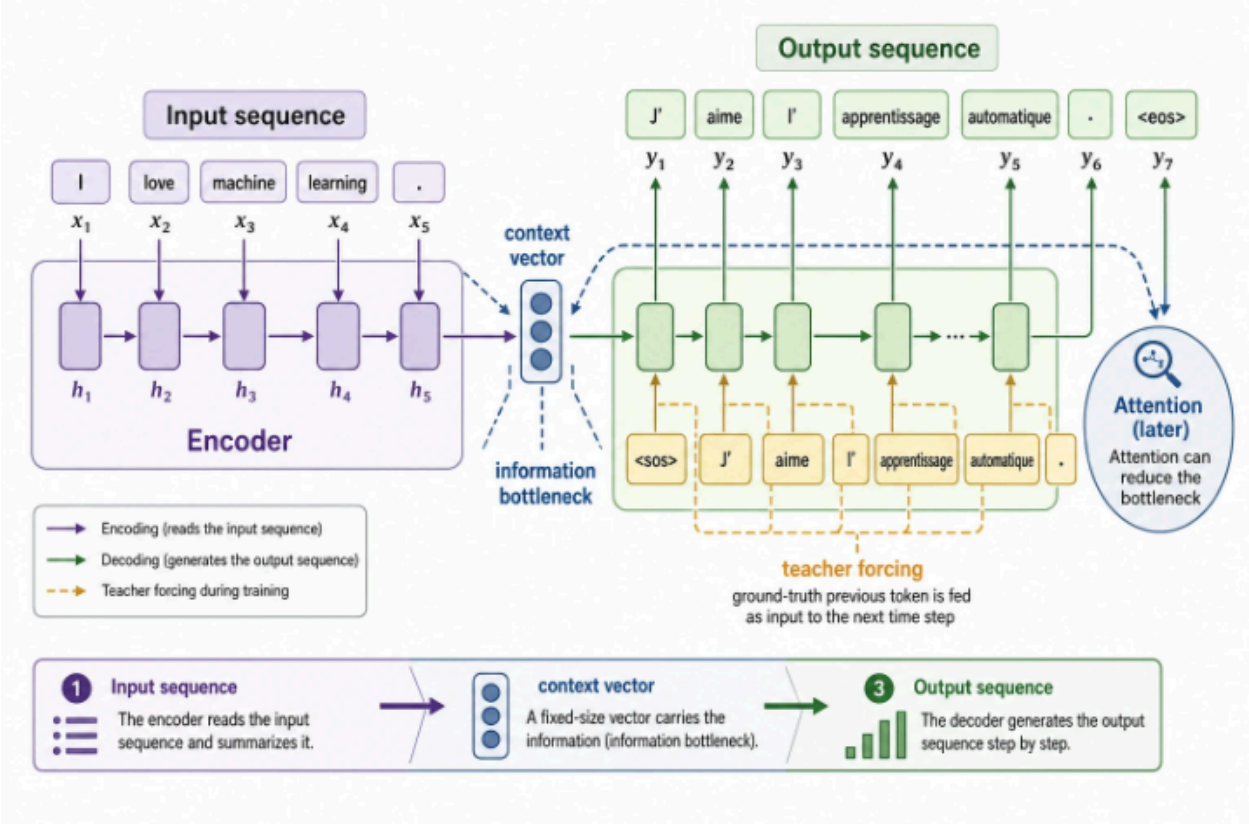
Main components:

1. Encoder

The encoder processes the input sequence and converts it into a representation that captures information from the input.

2. Decoder

The decoder uses the encoded representation to generate the output sequence, usually one token at a time.



Example: Machine Translation

Input sentence:

```
How are you?
```

The encoder processes the input sentence and creates a representation.

The decoder uses that representation to generate the translated sentence:

```
Comment allez-vous ?
```

Applications:

- Machine translation.

- Text summarisation.
- Text generation.
- Conversational systems.
- Speech recognition.

Limitation of the traditional encoder–decoder architecture:

In early RNN-based Seq2Seq models, the encoder often compressed the entire input sequence into a fixed-size context vector.

For long sequences, this could cause information loss.

The attention mechanism was introduced to help the decoder access relevant information from different encoder time steps.

Key Point: The encoder converts the input sequence into a representation, and the decoder uses that representation to generate the output sequence.

89. What is the attention mechanism, and why was it introduced?

The attention mechanism is a technique that allows a neural network to focus on the most relevant parts of the input when generating an output.

It was introduced to overcome limitations in traditional sequence-to-sequence models, particularly the difficulty of representing long input sequences using a single fixed-size context vector.

Example: Machine Translation

Consider the input sentence:

`The student is reading a book.`

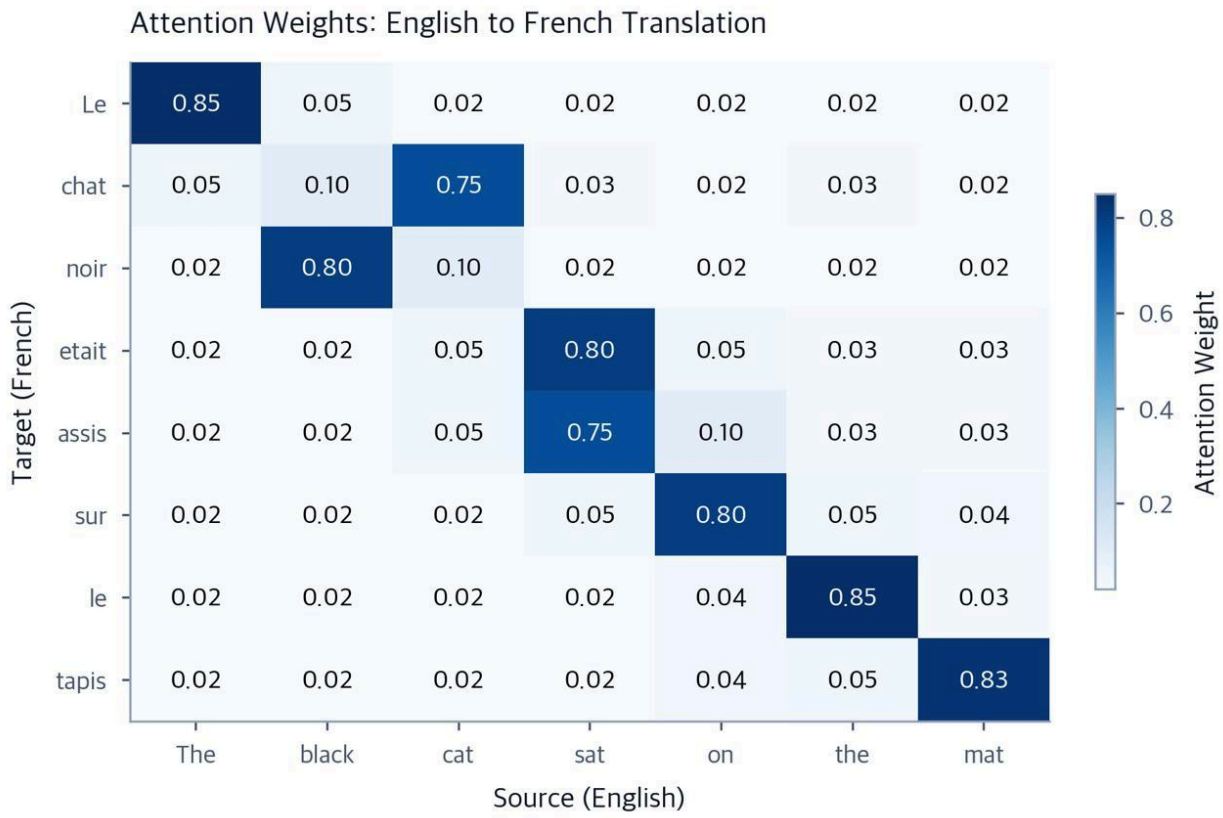
When translating the word `student`, the decoder should focus more on the corresponding word in the input rather than treating every input word as equally important.

Attention helps the model assign different importance weights to different input positions.

How does attention work?

1. The model calculates relevance scores between the current decoding state and encoder representations.
2. It converts these scores into attention weights, commonly using softmax.
3. It combines the encoder representations using those weights.

4. The resulting context information helps generate the next output token.



Why is attention important?

- Helps models focus on relevant input information.
- Improves handling of long sequences.
- Reduces dependence on a single fixed-size context vector.
- Helps capture relationships between different parts of a sequence.

Connection to Transformers:

The Transformer architecture is built around attention mechanisms rather than recurrent processing. Its self-attention mechanism allows tokens to interact with other tokens in the input sequence.

Key Point: Attention allows a model to focus on relevant parts of the input instead of relying on a single fixed-size representation.

90. How would you build a sentiment classification model using Deep Learning?

Sentiment classification is an NLP task that identifies the emotional tone or opinion expressed in text.

A sentiment classification model can classify text into categories such as:

- Positive
- Negative
- Neutral

For example:

```
"The service was excellent."
```

Expected output:

```
Positive
```

To build this model, I would follow these steps.

Step 1: Collect the Dataset

Collect text samples with their corresponding sentiment labels.

Example:

Text	Sentiment
The product is excellent	Positive
The service was terrible	Negative
The product is okay	Neutral
I really enjoyed it	Positive

Step 2: Preprocess the Text

Prepare the text before passing it to the neural network.

Common operations include:

- Removing unnecessary characters.
- Handling missing values.
- Normalising text where appropriate.
- Tokenisation.
- Converting tokens into numerical sequences.
- Padding sequences to a consistent length.

Step 3: Convert Text into Numerical Representations

Use a tokeniser to convert words into numerical IDs.

For example:

```
None
from tensorflow.keras.preprocessing.text import Tokenizer
from tensorflow.keras.utils import pad_sequences

texts = [
    "The product is excellent",
    "The service was terrible",
    "I really enjoyed it"
]

tokenizer = Tokenizer(num_words=5000)

tokenizer.fit_on_texts(texts)

sequences = tokenizer.texts_to_sequences(texts)

padded = pad_sequences(
    sequences,
    maxlen=10,
    padding="post"
)
```

Step 4: Build the Deep Learning Model

For a beginner-friendly implementation, we can use an Embedding layer followed by an LSTM layer and a classification layer.

```
None
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import (
    Embedding,
    LSTM,
    Dense
)

model = Sequential([
    Embedding(
        input_dim=5000,
        output_dim=64,
        input_length=10
    ),

    LSTM(64),

    Dense(3, activation="softmax")
])

model.compile(
    optimizer="adam",
    loss="sparse_categorical_crossentropy",
    metrics=["accuracy"]
)

model.summary()
```

Here:

- Embedding: Converts token IDs into dense vectors.
- LSTM: Learns sequential patterns from the text.
- Dense: Produces classification scores.
- Softmax: Converts scores into class probabilities.

Step 5: Train the Model

Split the dataset into training and testing sets, with a separate validation set where appropriate.

```
None
model.fit(
    X_train,
    y_train,
    validation_data=(X_val, y_val),
    epochs=10,
    batch_size=32
)
```

Step 6: Evaluate the Model

Evaluate its performance using:

- Accuracy.
- Precision.
- Recall.
- F1-score.
- Confusion matrix.

Step 7: Make Predictions

```
None
predictions = model.predict(X_test)

predicted_classes = predictions.argmax(axis=1)
```

The predicted class represents the sentiment category.

Important: The example assumes three classes encoded as integers `0`, `1` and `2`, with `X_train`, `y_train`, `X_val` and `y_val` already prepared. In a real project, the tokenizer must be fitted on training text only, and the class-to-label mapping must be maintained.

Key Point: A Deep Learning sentiment classification model uses text preprocessing, tokenisation, embeddings and neural network layers such as LSTM to classify text into sentiment categories.